

Development of an Energy-Optimized Control Allocation for Overactuated Omnidirectional Airships

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Kurzfassung

Eine unbemanntes, omnidirektionale Luftschiffdrohne bietet einzigartige Vorteile und Anwendungsmöglichkeiten in verschiedenen Bereichen. Die Tatsache, dass das Luftschiff unbemannt ist, eröffnet neue Möglichkeiten in Bezug auf Automatisierung und Autonomie. Es kann über längere Zeiträume in der Luft bleiben, ohne aktiv von Pilot gesteuert zu werden, was seine Nützlichkeit für Aufgaben wie Kartierung, Datenerfassung erheblich erhöht.

Diese Masterarbeit konzentriert sich auf ein omnidirektionales unbemanntes Luftschiff, das sowohl mit 4 auslenkbaren Rudern als auch mit einem überaktuierten Antriebssystem bestehend aus 9 Motoren ausgestattet ist. Der Hauptschwerpunkt liegt dabei auf der Untersuchung von Aerodynamik der Finnen zusammen mit ihren auslenkbaren Rudern, die durch Ausnutzung der Aerodynamik eine effiziente Ansteuerung im Flug ermöglichen. Die jetzige Ansteuerung der Luftschiff-Drohne berücksichtigt nur die aktiven Komponenten des Antriebssystems. Ziel ist es, die Ruder in die Ansteuerung mitzuintegrieren, um die Energieeffizienz des Luftschiffs zu verbessern und dadurch längere Flugdauern zu ermöglichen.

Zu diesem Zweck wird das Strömungsverhalten des NACA0015-Profiles mit verschiedenen Ruderverstellungen genauer untersucht. Es wird die Beziehung zwischen Anstellwinkel (AoA), Fluggeschwindigkeit, Ruderablenkung und den resultierenden aerodynamischen Kräften modelliert, was die Grundlage für den Rudersteueralgorithmus im Flugregelung bildet. Dieser ermöglicht es dem Flight-Controller, die optimale Ruderablenkung basierend auf den aktuellen Flugbedingungen zu ermitteln. Die Möglichkeit einer Zusammenarbeit von auslenkbaren Rudern und Seitenantrieben wird untersucht und entsprechend umgesetzt. Die funktionalen Algorithmen und die aerodynamische Effektivität der Finnen und Ruder werden in der Simulationsumgebung Gazebo evaluiert und analysiert.

Abstract

An unmanned omnidirectional airship offers unique advantages and applications in various fields. The fact that the airship is unmanned opens up new possibilities for automation and autonomy. It can remain airborne for extended periods of time without needing to be piloted, greatly increasing its utility for tasks such as mapping, data collection and logistical transport. This Master's thesis focuses on an omnidirectional unmanned airship which is equipped with both aerodynamic fin actuators and electric thrusters. The main focus of this work is the study of fins with deflectable rudders, which are designed to generate aerodynamic forces to reduce the load on the side thrusters. The aim is to optimise the use of these rudders to improve the energy efficiency of the airship, thereby enabling longer flight duration. For this purpose, the aerodynamics of the airfoil NACA0015 with various deflecting edges is taken into closer study. The relationship between angle of attack (AoA), airspeed, rudder deflection and the resulting aerodynamic forces is modelled, which forms the basis for the rudder control algorithm in PX4, allowing the system to determine the optimum deflection based on the current flight conditions. The possibility of cooperation between fins with deflectable rudders and side thrusters is being examined and implemented. The functional algorithms and aerodynamic effectiveness of the fins and rudders will be evaluated and analysed in a simulation environment Gazebo.

Contents

Erklärung	iii
Kurzfassung / Abstract	v
Abbildungsverzeichnis	ix
Tabellenverzeichnis	xi
1 Introduction	1
1.1 Approach and Structure of the Thesis	2
1.2 Introduction of the airship Looncraft Squid	2
2 Fundamentals	5
2.1 Dynamics of the Airship	5
2.2 Fundamentals properties about wing's Design	8
2.2.1 Fundamentals about Airfoil	9
2.2.2 NACA Airfoil and thin airfoil theory	11
2.2.3 Dynamics of an Airfoil	13
2.2.4 Reynolds Number (Re)	14
2.2.5 Aerodynamic Forces of an Airfoil	14
2.2.6 Lift, drag, and moment of an Airfoil	15
2.2.7 Trailing Edge Flap	20
3 Methodology and Approach of nondimensional coefficient modelling	22
3.1 Analytical approach	22
3.2 Numerical Approach	29
3.2.1 Introduction of Xflr5	29
3.2.2 Non-Dimensional Coefficient Estimation using xflr5	30
3.3 Development of a Parameter List for Control Allocation	36
3.3.1 Neglecting of Pitch Moment	36
3.3.2 Approximation of Cl and Cd	37
3.3.3 Approximation of Cn and Ca	37
3.3.4 Parameterisation of Deflection	40
3.3.5 Stall Boundries	41

Contents

3.3.6	Response Outside the Operating Range	42
3.3.7	Methodology for LookUp Table Retrival	43
3.3.8	Design and Integration of Rudder's Control Allocation	45
3.4	Establishing and Configuration of the Simulation Environment	46
4	Evaluation	48
4.1	Interpretation and Evaluation of Xflr5 Results	48
4.1.1	Symmetry between FlapUp and FlapDown curves	48
4.1.2	Aerodynamic in low speed range	49
4.1.3	Evaluation of Approximation of C_n	51
4.2	Validation of the Implementation in PX4	53
4.3	Proof of the Negligibility of Coefficient C_a	56
4.4	Evaluation of flight performance	59
4.5	Trajectory comparison	60
5	Conclusion	64
	Literaturverzeichnis	67

List of Figures

1.1	The airship "Looncraft Squid" from Roboloon, taken from [3]	3
1.2	Demonstration of the body frame of the airship, with one coordinate system at the centre of volume (CoV) and the other at the centre of gravity	3
2.1	Fin's coordinate system in "+"- and "x"-configuration	7
2.2	Schematic of the 9 motor forces, taken from [3]	8
2.3	Illustration of dihedral, tapered and sweep wing	10
2.4	Illustration of an airfoil showing the cross-sectional shape, leading edge, trailing edge, chord line, and camber line	11
2.5	NACA 4 digit airfoil specification, based on the example of NACA0015	12
2.6	Illustration of a NACA0015 profile, generated in Xflr5	13
2.7	Illustration of an airfoil showing the lift l , drag d and moment m and airspeed V_∞	16
2.8	Illustration of unit span l and unit area S	18
2.9	Common shapes of lift and drag curves with clearly identifiable stall angle α_{stall}	19
2.10	Various types of trailing edge flap. [14]	21
3.1	Distribution of the pressure coefficient C_p each of upper and lower surface, using the example of NACA0012. [4, figure 5.11]	24
3.2	η_d for plain flap, [12, Fig. 10]	27
3.3	3-D effectiveness factor by $\tau = 0.5$	27
3.4	2D- and 3D-Panel Method	30
3.5	Deflected airfoil shapes creating in Xflr5's "Direct Airfoil Design tool".	32
3.6	Airfoil analysis in Xflr5	32
3.7	Fin area shape comparison	33
3.8	curve family of $C_l(\alpha)$, each family represent one of the airspeed and Reynolds Number listed in table 3.2.2	34
3.9	curve family of $C_l(\alpha)$, each family represent one of the airspeed and Reynolds Number listed in table 3.2.2	35
3.10	Simplification	38
3.11	curve family of $C_n(\alpha)$, each family represent one of the airspeed and Reynolds Number listed in table 3.2.2	39

List of Figures

3.12	Approximation of C_n limited in laminar flow. The result is equivalent to the operating range of deflectable rudder's.	41
3.13	Extended operating area of C_n	42
3.14	Looking for the value of C_n by given airspeed, AoA and rudder's deflection.	43
3.15	Looking for the appropriate rudder's deflection by given airspeed, AoA and rudder's deflection.	44
3.16	The system's control loop which is extended by a rudder's deflection controller, which is placed before the control allocator for the thrusters.	45
4.1	curve family of $C_n(\alpha)$ for airspeed under 10km/h	50
4.2	Fit parametric coefficients of $C_n(\alpha)$, where the slop m is fitted using 1st-order polynomial and the offset b is fitted using 2nc order polynomial	52
4.3	Comparison of C_n between Px4 and Gazebo respectively to each fin.	54
4.4	Comparison of F_n between Px4 and Gazebo respectively to each fin.	55
4.5	Comparison between C_n and C_a of each fin.	57
4.6	Comparison between F_n and F_a of each fin	58
4.7	Predefined trajectory waypoints in the simulation for guiding the airship	59
4.8	Comparison of the recorded trajectory and altitudes from both runs in Gazebo simulation	60
4.9	Comparison of allocating torques for the run with both side thrusters and rudder deflecton	62
4.10	Comparison of allocating torques for the run with only side thrusters	63

List of Tables

1.1	The key data points of the airship "Looncraft Squid"	4
3.1	Considered deflection grip points in -45° to 45°	33
3.2	Considered velocity and Reynolds Number	33

1 Introduction

Unmanned Aerial Vehicles (UAVs) has gained attention over the last decade due to a number of advantages, including cost-effectiveness and versatility [7]. VTOL (Vertical Takeoff and Landing) capability in UAVs allows aircraft to take off, hover and land vertically. VTOL UAVs is able to be deployed in almost any location, including urban and rural environments with complex terrain, making them ideal for operations in confined or remote areas where traditional runways are not available, hence VTOL UAVs are suitable for a wide range of missions. The ability to hover and perform precise movements is crucial for applications like surveillance, inspection and search applications. [18]

VTOL UAVs come in different types and configurations, depending on requirements of specific applications. The multi-rotor design arranges several rotors in a pattern for propulsion and directional changes. One of the representatives is the quadcopter which is compact, lightweight, easy to use and highly agile with rapid directional changes [6]. Airship drones is another type of VTOL UAV, which belongs to the lighter-than-air (LTA) category that relies on buoyant gases stored in the airship's envelope to compensate the gravity and provide lift. This principle allows the vehicle to hover and stay aloft for long periods of time, as lift doesn't consume energy. [13][8]

An omnidirectional equipped with multiple thrusters and deflectable rudders, omnidirectional airships move forward, backward, laterally and vertically, making them highly versatile. Their advanced steering mechanisms and aerodynamic design enable precise control and stability. In contrast to quadcopters, omnidirectional airship acts with low noise and is more energy-efficient due to buoyant lift reducing the need for continuous motor thrust against gravity. With fin deflection control, it has the potential to further minimize energy consumption and extends fly time.

Given these advantages, there is a need to investigate the feasibility and effectiveness of fin deflection control in omnidirectional airships aiming to optimize their energy efficiency and operational capabilities. This proposal builds upon a preexisting platform developed in [3], where an omnidirectional airship is designed to move and change direction via onboard motors. The airship is extended with four rudders featuring deflectable fins. This research aims to explore and model the impact of fin deflection on an airship's performance, paving the way for enhanced airship designs that fully leverage this technology.

1.1 Approach and Structure of the Thesis

The aim of this work is to extend the control system of the airship to include rudder deflection in order to relieve the lateral drive and thus achieve greater energy efficiency. The approaches and procedures are as follows:

Step 1: Modelling of fin's Aerodynamic. The controller requires knowledge about the aerodynamics of the fin and the effect of deflecting rudder in order to be able to make algorithmic decisions. For this reason, the aerodynamics of the fin the effect of deflecting rudders is analysed in the first step of this work. A simplified aerodynamic model for the PX4 flight controller is to build up accordingly.

Step 2: Deflection Control. The dynamics of the airship are described by setting up the Newton-Euler equation for the fins. This equation is used to determine the appropriate deflection.

Step 3: Extension of the Control System. The rudder's deflection control of the fins is an extension to the preexisting control system. It is important to ensure that the rudder's deflection control and the control allocation of the side thrusters is able to cooperate well with together.

Step 4: Simulation and evaluation. The dynamic model of the deflectable rudders is analysed in the Gazebo simulation environment, together with the control allocation of the lateral drive. An environmental reference of the fin specific aerodynamics is required for the simulation. Run data from both PX4 and Gazebo are taken into account for the validating the performance of the rudder's deflectoin control.

The following chapters are based on the steps listed. Firstly, an introduction to the Luftshiff and the previous work follows. In Chapter 2, the basics of the aerodynamics of the fins are analysed. The implementations in steps 2 and 3 are described in Chapter 3. This is followed by an evaluation in Chapter 4. Finally, Chapter 5 reflects on the work once again.

1.2 Introduction of the airship Looncraft Squid

The sooncraft squid is an outdoor prototype constructed by roboloon in Stuttgart Germany. This prototype validates the novel propulsion concept and demonstrates the maneuverability with 6 degrees of freedom (DoF) and the ability to fly in challenging wind conditions. The airship is 8 meters long and the diameter is less than 2 meters. In order to gain more stability, the airship's Center of Gravity (COG) is deliberately positioned

1.2 Introduction of the airship *Looncraft Squid*

under the center of volume (COV) (Figure 1.2). The appearance of *Looncraft Squid* is depicted in Figure 1.1.

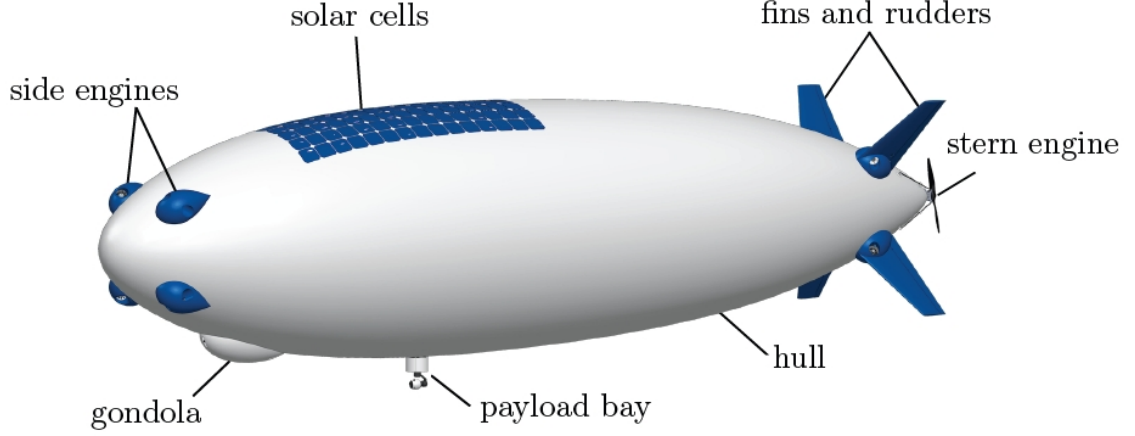


Figure 1.1: The airship "Looncraft Squid" from Roboloon, taken from [3]

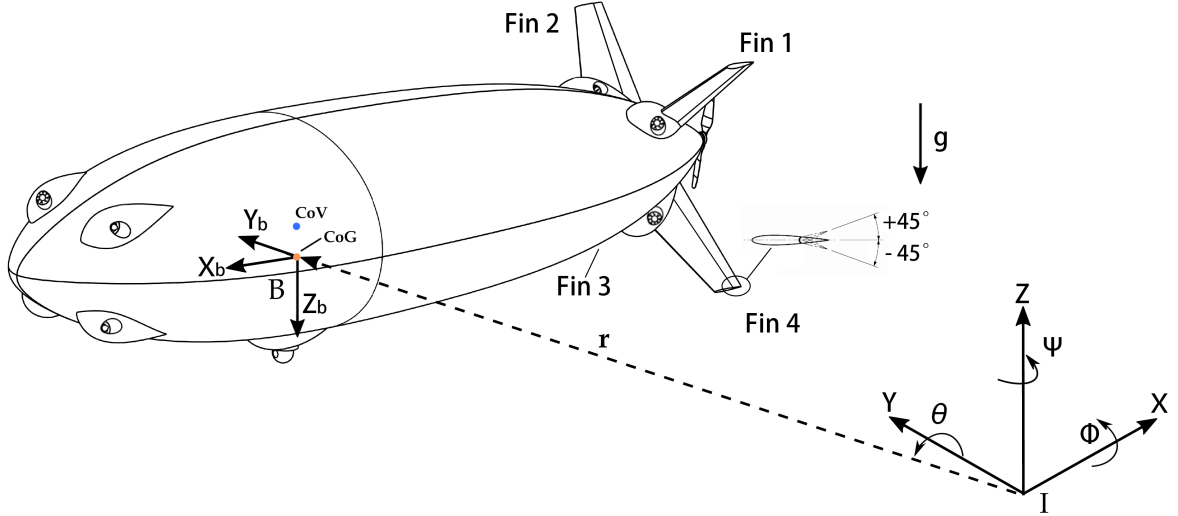


Figure 1.2: Demonstration of the body frame of the airship, with one coordinate system at the centre of volume (CoV) and the other at the centre of gravity

The airship is equipped with a total of nine thrusters: four side thrusters arranged in x-configuration at the front and other four in the same x-configuration at the rear and an additional stern-engine at the central rear. All of the eight side thrusters are ducted

1 Introduction

fans of the same type, each of them generates a unidirectional force and the fans are protected with a bowl-shaped housing. The stern motor generates thrust in the positive x_b -direction [3].

The airship has four rudders at the rear side, each rudder has a deflectable fin. The rudders are attached perpendicular to the bowl-shaped housing of those four side thrusters at the rear, therefore the rudders have the same x-configuration as the side thrusters and each rudder's chord is parallel to the airship's x_b -direction. All the nine thrusters and four rudders serve as actuators which are available for control command.

Key Parameter	Value	Unit
Length	8	m
Diameters	2	m
Mass	17	kg
Maneuverability	6	DoF
Propulsion	9	thrusters
Rudders	4	[-]
Max. wind speed	25	km/h
Fin surface area	0.222	m ²
Fin average chord	0.331	m ²

Table 1.1: The key data points of the airship "Looncraft Squid"

2 Fundamentals

This chapter lays the groundwork for the comprehensive modelling of the airship's dynamic and the rudder's aerodynamic. It delves into the essential concepts and theories that form the basis of this research. By providing a thorough understanding of these foundational elements, solid framework upon which the subsequent chapters will build.

2.1 Dynamics of the Airship

For airships, The terms “rigid” and “non-rigid” refer to the structural design and indicate whether an airship has a structural framework inside the envelope, while “semi-rigid” incorporates a partial internal framework that helps maintain shape and structural integrity. [10]

The Looncraft Squid is a non-rigid airship with no internal frame, so deformation due to weight and thrust must be taken into account and compensated for. However, the computational resources of the flight controller on board are limited, so it is necessary to keep the dynamics of the airship as simple as possible. Due to the assumption, that the hull has sufficient inner pressure and has no contact with other rigid objects, the effects of deformation on the dynamics of the airship are neglected in [3] and the airship is considered as a rigid body. This assumption is used throughout this paper, unless otherwise stated.

For conventional reasons, the body-fixed coordinate system of the airship with origin in CoV (denoted as “B-Frame”) and the body fixed coordinate system with origin in fin's area center point (denoted as “F-Frame”) are used in this proposal. Since there is no possibility to measure the airspeed of the fin, the fin's air speed needs to be derived from the air speed of the airship, measured in coordinate system with origin in CoV. The Transformation of the air speed from coordinate system with origin in CoV and to i-th fin's coordinate system is written as following:

$${}_B\mathbf{V}_{\infty,i} = {}_B\mathbf{V}_{\infty} + \boldsymbol{\omega} \times {}_B\mathbf{r}_{CoV,i} \quad (2.1)$$

2 Fundamentals

$${}_F\mathbf{V}_{\infty,i} = \mathbf{T}_1 \times {}_B\mathbf{V}_{\infty,i} \quad (2.2)$$

with ω the angular velocity between the two coordinate system; ${}_B\mathbf{r}_{CoV,i}$ the distance from CoV to i-th fin's coordinate system and T_1 the rotation matrix around x-axis.

The airship Looncraft Squid has 9 propulsion thrusters and 4 deflectable rudders. The control vector $\underline{u}$ is written respectively as:

$$\underline{u}_{act_{thruster}} = [u_1 \quad u_2 \quad \dots \quad u_9] \quad (2.3)$$

$$\underline{u}_{act_{fin}} = [u_{fin_1} \quad u_{fin_2} \quad u_{fin_3} \quad u_{fin_4}] \quad (2.4)$$

where u_1 to u_9 denotes the control for the nine thrusters, which can be described entirely by the motor's angular velocity: $u_{fin_i} = \omega_i^2$ for $i \in [1, \dots, 9]$ and u_{fin_1} to u_{fin_4} denotes the control for the 4 deflectable rudders: $u_{fin_i} = \delta_i$ for $i \in [1, \dots, 4]$ and δ the deflection angle of the rudder, currently with a maximum of tiltable range from -45° to 45° . In contrary, the motion has only 6 degrees of freedom: the translational motion in x-, y-, and z-direction and the rotation around x-, y-, and y-axis which are the roll angle ϕ , pitch angle θ and yawl angle ψ .

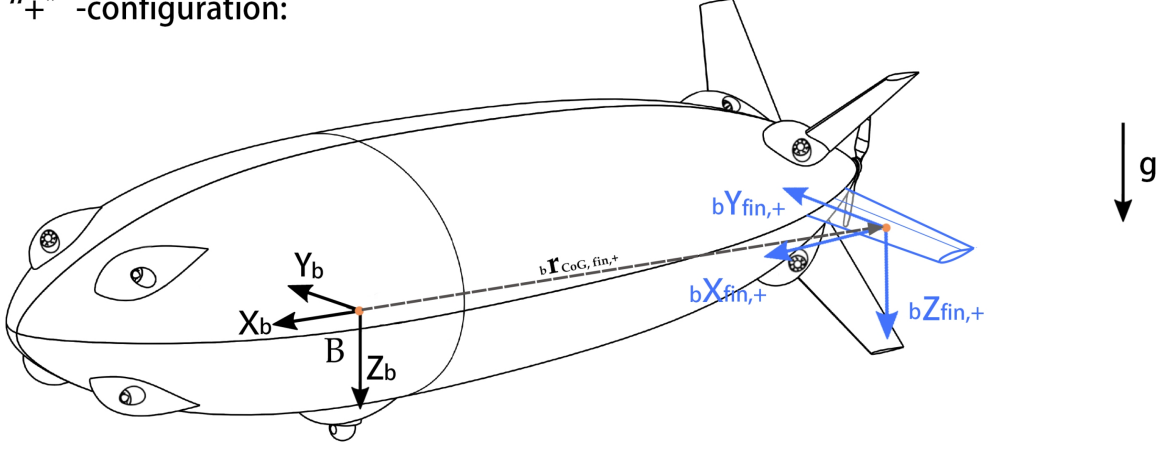
$$\underline{y}_{act} = [x \quad y \quad z \quad \phi \quad \theta \quad \psi] \quad (2.5)$$

The velocity and angular acceleration is simply the first and second derivatives of the state $\underline{y}$: $\dot{\underline{y}}$ and $\ddot{\underline{y}}$.

The dynamic of the airship is modeled by the Newton-Euler-Equation. In the preliminary work of Pala [3], the mass matrix $\underline{\underline{M}} \in \mathbb{R}^{6 \times 6}$ is defined as a block diagonal matrix with upper left diagonal block being the masses effecting on the COG in diagonal and the identity matrix in the lower right diagonal block. Vector $\underline{k}$ denotes the internal force which refers to the Coriolis and centripetal forces. The right side $\underline{q}$ denotes the external forces and torques, mainly composed of gravitational force F_g , buoyancy force F_b and aerodynamical forces caused by the airship's actuators. The forces and torques created by the thrusters is modeled by $\underline{\underline{B}}_{act} \cdot \underline{u}_{act}$, with $\underline{u}_{act}$ the applied command to the actuators u_i and $\underline{\underline{B}}_{act}$ the effectiveness matrix. The dynamic of the airship with consideration of the actuators is given in general by:

$$\underline{\underline{M}}(y, t) \ddot{\underline{y}} + \underline{k}(y, \dot{y}, t) = \underline{q}(y, t) + \underline{\underline{B}}_{act} \cdot \underline{u}_{act} \quad (2.6)$$

“+” -configuration:



“x” -configuration:

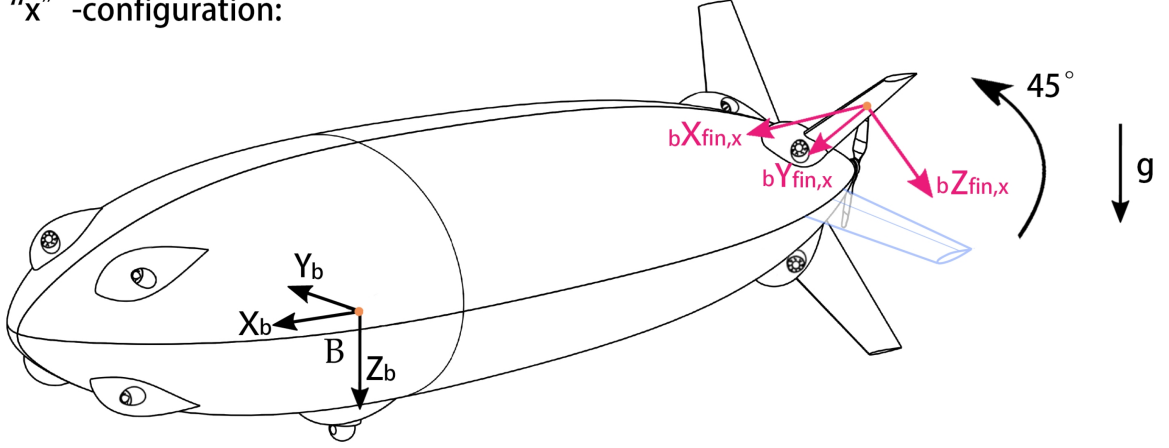


Figure 2.1: Fin's coordinate system in “+”- and “x”-configuration

The nine thrusters generate unidirectional forces forces are depicted as F_{BUL} , F_{BUR} , F_{BLR} , F_{BLL} which act at the front side of the airship, F_{SUL} , F_{SUR} , F_{SLR} , F_{SLL} which act at the back and F_{SP} the stern motor generate a thrust at the very end. Additionally, aerodynamic forces generated by the fins and deflecting rudders are demonstrated in Form of normal forces which is perpendicular to the fin's surface. These forces are denoted as $F_{n,fin1} \dots F_{n,fin1}$. The mentioned forces and each of the orientation are demonstrated in Figure 2.2.

2 Fundamentals

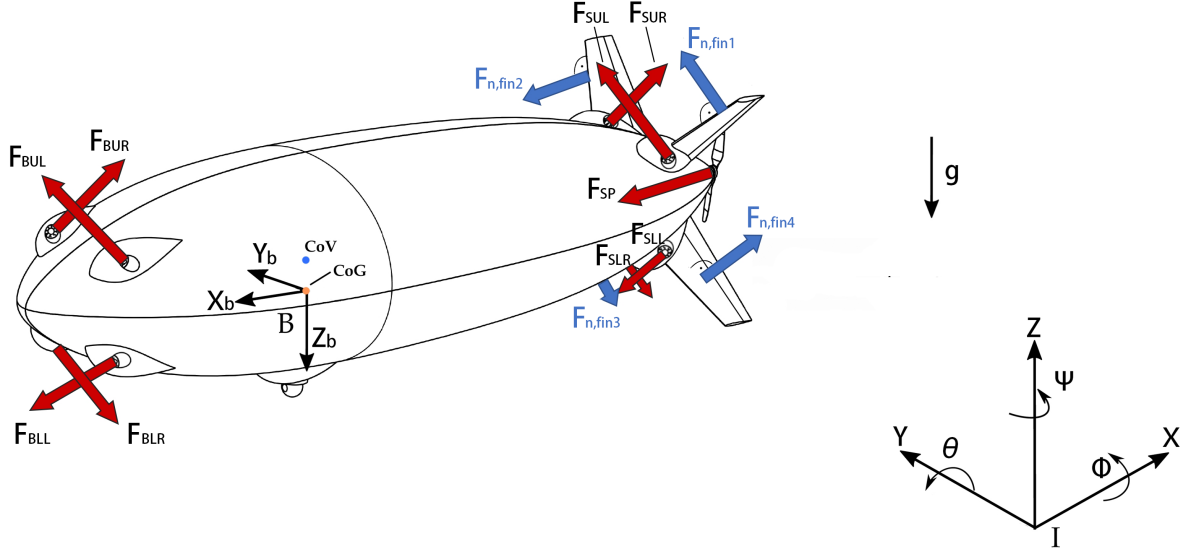


Figure 2.2: Schematic of the 9 motor forces, taken from [3]

When discussing about the fin's dynamic, the Newton-Euler-Equation is simplified. First, due to the reduction of $\underline{y} = \underline{y}_{fin}$, the mass matrix $\underline{\underline{M}}$ is reduced to the matrix block at the bottom right, which is equal to an identity matrix I . The internal force $\underline{k}$ is irrelevant for the fin's dynamic, therefore neglected. The external force is due to the rudder's deflection $u_{act,fin}$. As the fins not being responsible for translational movement, the airship's state $\underline{y}$ is reduced to those angular variable. For rudder's deflection control, the angular acceleration is $\underline{\Omega}$ given as the set point for the rudder's deflection control. The vector for the airship's state written in B-Frame is given by:

$$\ddot{\underline{y}}_{fin} = \underline{\underline{B}}_{act,fin} \cdot \underline{u}_{act,fin} \quad (2.7)$$

with $\underline{y}_{fin} = (\phi, \theta, \psi)$.

2.2 Fundamentals properties about wing's Design

This section provides a brief overview of fundamental properties in wing design which are relevant to this proposal.

2.2 Fundamentals properties about wing's Design

Dihedral: Dihedral refers to the upward angles of the wings relative to the horizontal plane, so the tips are higher and the wings appear to be a V-shape when viewing from the front. The negative form of dihedral is called anhedral, and the dihedral angle is assigned to be γ . Dihedral wings affect the aircraft's roll motion and help stabilizing the aircraft from making Dutch roll, which is a combination roll, yaw and side slip. The x-configuration of the fins by the airship Looncraft Squid can be regarded as a combination of dihedral and anhedral wings with an angle $\gamma = 45^\circ$ respectively. [9]

Sweep: The plan view of a wing refers to its shape as viewed from above, which provides an outline of the wing's geometry. Swept-back wings can be recognized from the plan view where the wings of an aircraft are angled backward to the fuselage instead of being perpendicular to it. The swept back angle λ is measured by the imaginary line perpendicular to the hull's central line and the wing's leading edge. Swept-back wing can reduce bending loads and improved stiffness. It also helps minimizing wave drag and induced drag, although this phenomenon is mainly observed by high speed aircraft, but less pronounced in low speed condition. [4] [9]

Tapered: A tapered wing is characterized by a chord length that decreases along the span width from the wing root towards the wingtip. This design optimizes aerodynamic efficiency by reducing induced drag and improving lift distribution. According to the change in chord length across the span of a wing, the change of an aerodynamic parameter can be described as a spanwise distribution, like for example the spanwise distribution of chord $c(y)$ which varies along the wing's span due to differences in airfoil shape and size. Since the fins of the airship Looncraft Squid have both straight leading and trailing edges with same airfoil shape across the span width, the straight-tapered wing is relevant for this proposal. The tapered degree is characterized by ratio

$$\lambda = \frac{c_r}{c_t} \quad (2.8)$$

with c_r the root's chord length and c_t the tip's chord length. The main benefits for using tapered wing is to reduce the root bending moments and the straight tapered wing is easier to manufacture.

2.2.1 Fundamentals about Airfoil

An airfoil, also known as a wing profile, is a cross-sectional shape of a wing with its plane perpendicular to the wing. It is a fundamental component of aircraft's wings, propeller, and other aerodynamic structures since the shape has major influence to the lift and drag [11]. It has mostly a curved streamlined shape to produce lift when it interacts with a fluid, typically air, since air wing with curved surface provides more lift than flat surface. Some terminologies are essential for characterizing an airfoil

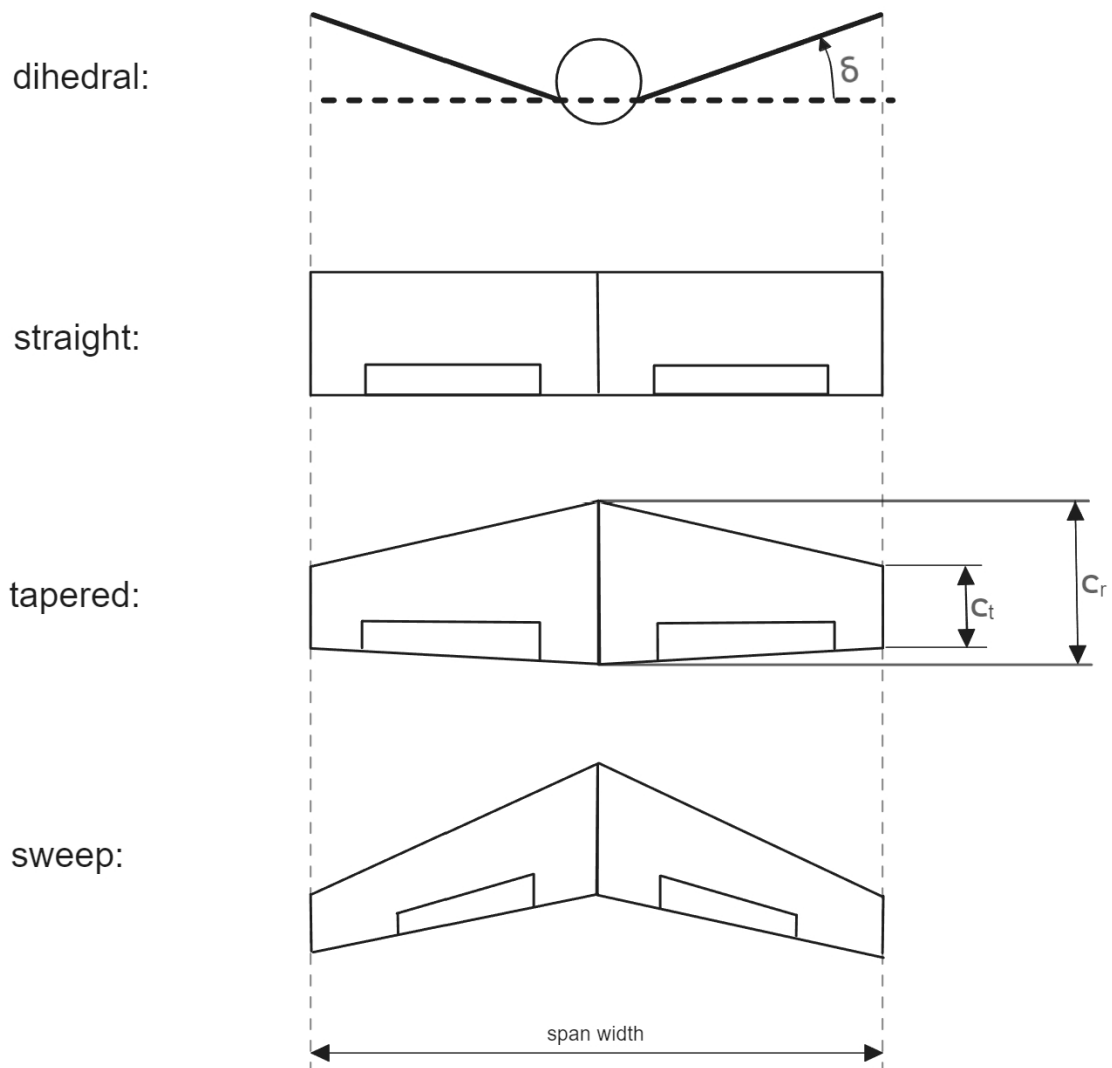


Figure 2.3: Illustration of dihedral, tapered and sweep wing

Camber: The curvature of the upper and lower surfaces of the wing is referred as camber. The camber is a measure of how much the airfoil deviates from being flat. An airfoil's camber can be positive, negative or zero. If an airfoil's upper surface is more curved than lower surface, the camber is positive; if the lower surface is more curved than the upper surface, the camber is negative. If the upper and lower surfaces have the same curvature or are equally flat, the airfoil has zero camber.

Chord: The chord c is another property that is essential for characterising the aerody-

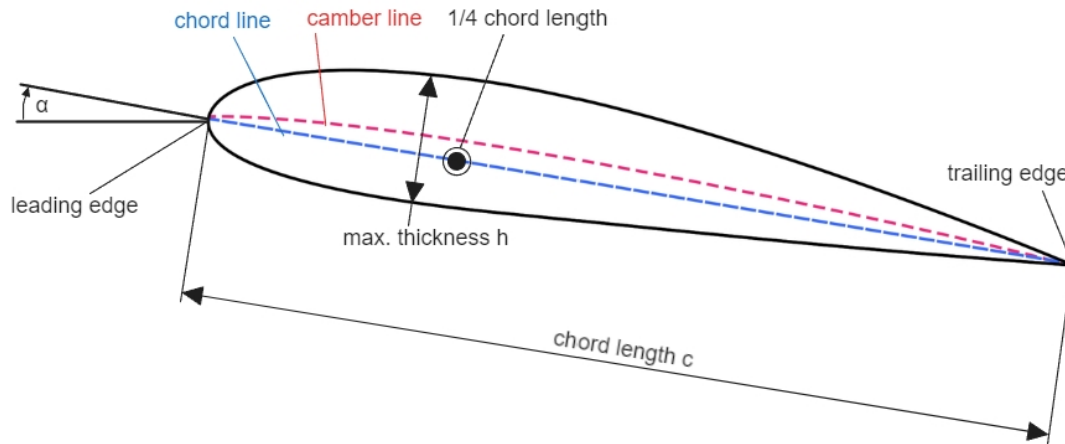


Figure 2.4: Illustration of an airfoil showing the cross-sectional shape, leading edge, trailing edge, chord line, and camber line

dynamic properties of the wing and airfoil. The chord is the straight line measured from the leading edge to the trailing edge of the airfoil. It characterises the width of the wing or airfoil in relation to the actual cross section, otherwise the chord length may vary along the span of the wing in the case of complex wing designs where the width of the wing is not constant. At zero camber, the distance from the top and bottom surfaces to the chord line should be equal, so zero camber implies symmetry of the airfoil.

Thickness: The thickness of an airfoil is the distance between its top and bottom surfaces, measured perpendicular to the chord line. It is usually given as a percentage of the chord length, but the thickness distribution along the chord line can vary due to the curvature design.[4][5]

2.2.2 NACA Airfoil and thin airfoil theory

In the field of aerodynamics, airfoils are standardised to provide a systematic and universally recognised framework for the design and analysis of aircraft wings. These airfoils are characterised by specific geometric parameters that define their shape and performance characteristics. A prominent set of standardised airfoils are the National Advisory Committee for Aeronautics (NACA) airfoils, developed by the forerunner of NASA in the 20th century. NACA airfoils are identified by a four-digit designation, with each digit providing specific information about the airfoil's characteristics. Here is a specification of the NACA 4-digit airfoil, with the chord measured from the leading edge:

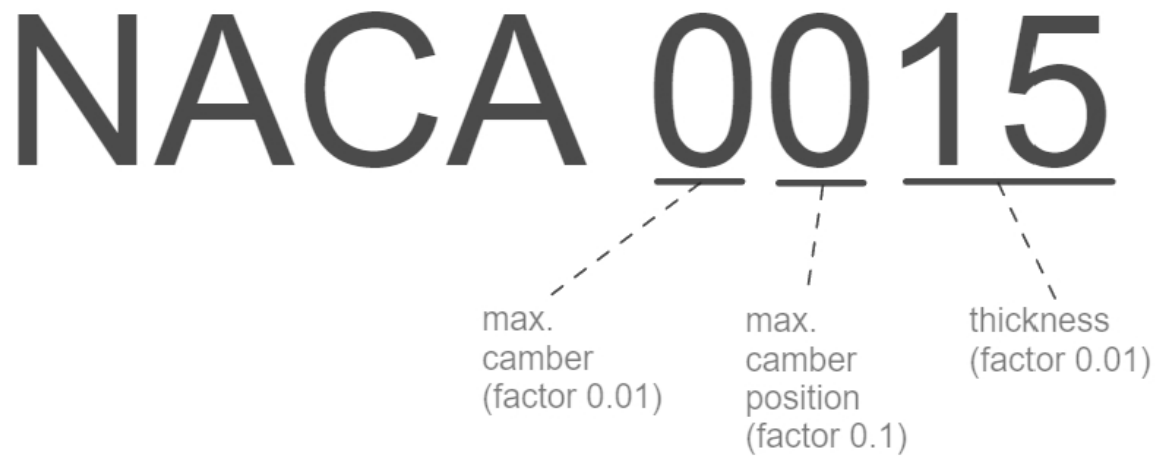


Figure 2.5: NACA 4 digit airfoil specification, based on the example of NACA0015

The first digit indicates the maximum camber of the airfoil, expressed as a percentage of the chord length. For example, if the first digit is 3, this means that the airfoil has a maximum camber of 3%. The second digit indicates the position of the maximum camber along the chord divided by 10. For example, if the second digit is 4, the maximum camber is at 40% of the chord. The last two digits express the maximum ratio of thickness to chord. The position of the maximum is defined at 30% of the chord. [1]

Thin airfoil theory is commonly used for various NACA four-digit airfoils. Thin airfoil theory is a simplified mathematical framework that assumes an airfoil generates lift primarily due to pressure differences between its upper and lower surfaces, neglecting effects such as viscous forces and three-dimensional flow. As the name suggests, this theory is suitable for thin airfoils, where the ratio of chord length to chord width must be significantly less than 1: $\frac{h}{c} \ll 1$; it is suitable for airfoils with small camber or even symmetry. Thin airfoil theory is used to analyse and predict lift and drag, especially at low AoA.

The airship Looncraft Squid has for wings of the airfoil profile NACA 0015. According to the NACA airfoil designation, this profile has zero camber, which means that the mean camber line is a straight line and the airfoil is symmetrical. The second digit is also 0, meaning that the maximum camber is at the midpoint along the chord. However, as the airfoil has zero camber, this parameter is less relevant. It is used for consistency with the NACA four digit system. The last two digits indicate that the airfoil has a thickness to chord ratio of 15% at 30% chord.

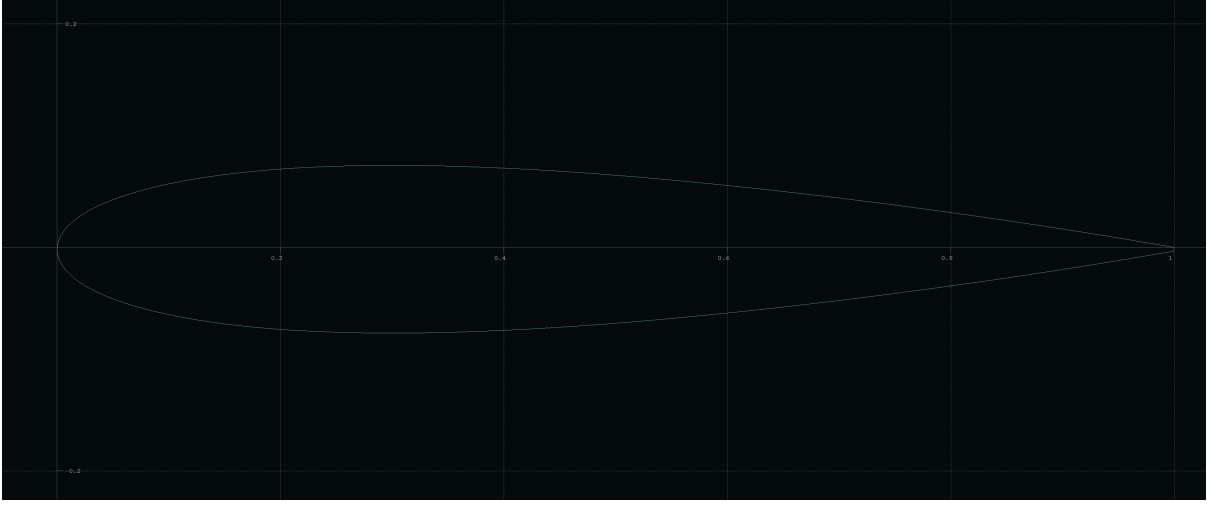


Figure 2.6: Illustration of a NACA0015 profile, genereated in Xflr5

2.2.3 Dynamics of an Airfoil

The stream velocity V_∞ , also called true airspeed, is a parameter which plays a significant role in context of aerodynamic. It can be referred as the speed and direction of the undisturbed freestream air through which the airfoil is moving. Conversely, it is interpreted as the speed of the air mass that is flowing over the airfoil [9]. The stream velocity is relative to the ground, the relation between stream velocity, wind velocity and the ground velocity of the airship is given as:

$$V_\infty = V_{ground} + V_{wind} \quad (2.9)$$

Another coordinate system commonly used in relation to flow velocity is the aerodynamic coordinate system (A-frame). It is often used to describe the aerodynamic forces and moments acting on an aerodynamic body. In this system the X-axis points in the direction of the flow velocity, the Y-axis is perpendicular to the flow and the Z-axis points towards the earth. The "Angle of Attack" (AoA, also denoted by " α ") is the angle between the chord line of an airfoil and the oncoming air, which is also the direction of the airflow's velocity. The AoA is a crucial parameter in aerodynamics as it affects lift and drag forces. A moderate value of AoA helps to optimise lift, but if the angle is too high, it can lead to airflow separation and stall. The AoA where stall appears is denoted as α_{stall}

2.2.4 Reynolds Number (Re)

The Reynolds number is a dimensionless quantity used in fluid dynamics and aerodynamics to characterize the relative importance of inertial forces to viscous forces within a fluid flow. The Reynolds number (Re) is defined by the following formula [2]:

$$Re = \frac{\rho \cdot V_{\infty} \cdot L}{\mu} \quad (2.10)$$

with:

ρ	density of the fluid, typically measured in $[kg/m^3]$
L	the mean chord width of an airfoil, measured in $[m]$
μ	dynamic viscosity of the fluid}
V_{∞}	the stream velocity relative to the subject, measured in $[m/s]$

The airship Loocraft Squid is considered to operate in a state where the flow velocity doesn't exceed 30km/h, which is extremely insignificant compared to the speed of sound. This velocity range results in relatively low Reynolds number values ($Re < 200,000$). At low Reynolds numbers, the flow around the wing/wing is characterised by laminar flow, which means less mixing and less vortices in the flow and less skin friction, so drag can be very low. In contrast, the streamlines of a turbulent flow break up and behave irregularly [4]. On the other hand, low Reynolds numbers can be a challenge for the control surfaces of the airship, as the rudders have low effectiveness. This issue will be further demonstrated and discussed in the Chapter 4.

2.2.5 Aerodynamic Forces of an Airfoil

As the wing moves through the free air stream, it generates a force R and a pitch moment. The force R can be divided into two components: the lift L , which is perpendicular to the direction of the flow, and the drag D , which is parallel to the direction of the flow.

According to Bernoulli's principle, the pressure near the wing surface decreases as the relative air speed around the wing surface increases, and vice versa. The leading edge of the wing divides the airflow into two streams along the upper and lower surfaces of the wing. Lift L is generated by the pressure difference between the upper and lower flows, depending on the shape of the airfoil and the current AoA. [9, Chapter 8, 10]

Drag can be broadly divided into two types: parasitic drag and induced drag. Parasitic drag, also known as profile drag, is primarily caused by the geometric shape of the wing

2.2 Fundamentals properties about wing's Design

which causes a drag by flow separation but also by skin friction due to the interaction between the airflow and the wing surface. Therefore, parasitic drag is not directly related to pressure as lift is. Induced drag, on the other hand, is associated with the production of lift and is related to pressure differentials and vortices at the wingtips, this part of drag is taken into account by transonic and subsonic speeds and is negligible for speeds far below than subsonic speed. The total drag D of an airfoil is the sum of skin friction drag D_f , pressure drag D_p due to flow separation and wave drag D_w which is regarded to be zero in this proposal [9, Chapter 9] [4, Chapter 5.12]:

$$D = D_f + D_p (+D_w) \quad (2.11)$$

The force vector written in aerodynamic coordinate is given by following:

$$\underline{R_A} = [-D \quad Q \quad -L] \quad (2.12)$$

With neglecting of shear stress Q for simplicity [4], the force R acting on an airfoil is given as:

$$\underline{R_A} = [-D \quad 0 \quad -L] \quad (2.13)$$

The pitch moment M is generated by the pressure difference and the displacement of the induced aerodynamic forces from the upper and lower surfaces respectively, as shown in figure 2.7. The resulting moment tends to rotate the airfoil around the pivot point, which is usually located at a quarter of the chord width ($c/4$) measured from the leading edge. [4]

2.2.6 Lift, drag, and moment of an Airfoil

The current AoA α has a significant effect on Lift L , drag D and the airfoil's pitch Moment M , but not only, Lift L , drag D and Moment M are also depending on further physical quantities:

- freestream velocity V_∞ , [m/s^2];
- freestream density ρ , [kg/m^3];
- area of the aerodynamic surface, which is denoted as wing area S , [m^2];

Dimensional analysis and the identification of dimensionless parameters is a method widely used in physics and engineering. The key idea underlying this approach is the

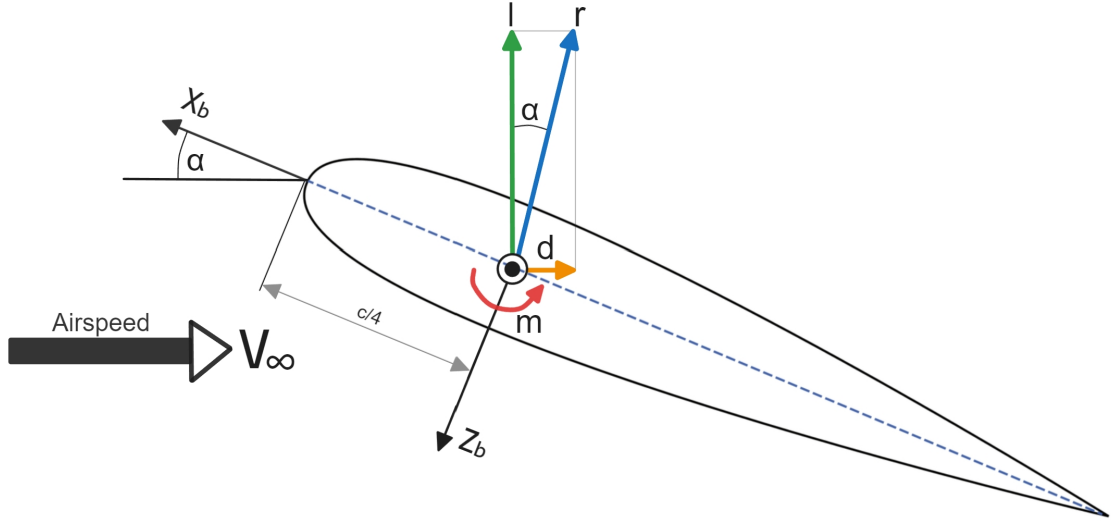


Figure 2.7: Illustration of an airfoil showing the lift l , drag d and moment m and airspeed V_∞

need for dimensional consistency on both sides of an equation. This methodology provides a systematic framework for analysing the relationships between physical quantities and the forces of lift, drag and moment exerted on a body moving through a fluid. The following derivation is cited from [9, Chapter 8] which offers a comprehensive derivation for modelling the lift, drag and moment. The distribution of pressure during the flight is essential in shaping the aerodynamic forces and moments acting on the wing.

The pressure is defined as force per unit area F/S where the unit area S is constructed by l^2 . The force can be expressed as the product of mass m and acceleration a . Ensuring dimensional consistency, the relationship among these quantities can be formulated as following [9, Chapter 8 (8-1)]:

$$F = p \cdot S = \frac{m}{l \cdot t^2} \cdot l^2 = \frac{ml}{t^2} \quad (2.14)$$

On the other side, the aerodynamic force is expressed as follows [9, Chapter 8 (8-2)]:

$$F = k\rho^a V^b l^c \quad (2.15)$$

where

2.2 Fundamentals properties about wing's Design

m:	mass
l:	unit length
t:	time
k:	unknown constant of proportionality
a, b, c:	exponents to be determined

Equalisation of the two forces and determination of the exponents by comparison of the exponents gives the following result [9, Chapter 8 (8-3)]:

$$\frac{ml}{t^2} = k \left(\frac{m}{l} \right)^a \left(\frac{l}{t} \right)^b l^c \quad (2.16)$$

with $a=1$, $b=2$ and $c=2$; k is chosen as $\frac{1}{2}C_r$ where C_r is a nondimensional coefficient relating AoA to the force and l^2 is replaced by the area S [9, Chapter 8 (8-4)(8-5)], a resultant force is obtained as follows:

$$r = \frac{1}{2} \rho V_\infty^2 S C_r \quad (2.17)$$

Lift and drag can be calculated by applying trigonometric function with AoA α to the resultant force r . When dealing with airfoils, area S is considered that of a wing of unit span as demonstrated in figure 2.8, this yields:

$$\begin{aligned} l &= r * \cos(\alpha) \\ d &= r * \sin(\alpha) \end{aligned} \quad (2.18)$$

When discussing the aerodynamics of an aircraft, the nondimensional coefficient is often taken into greater consideration than the dimensional coefficient. It allows the evaluation of aerodynamic properties independent of the geometric properties of the aircraft, so that the aerodynamics of different airfoils can be compared regardless of the actual airfoil size and flight speed. The non-dimensional coefficients, i.e. the airfoil sectional lift C_l , the airfoil sectional drag C_d , the airfoil quarter chord moment C_m , are introduced and the 3D lift L , 3D drag D and 3D pitch moment M in relation to the whole wing are given as follows:

2 Fundamentals

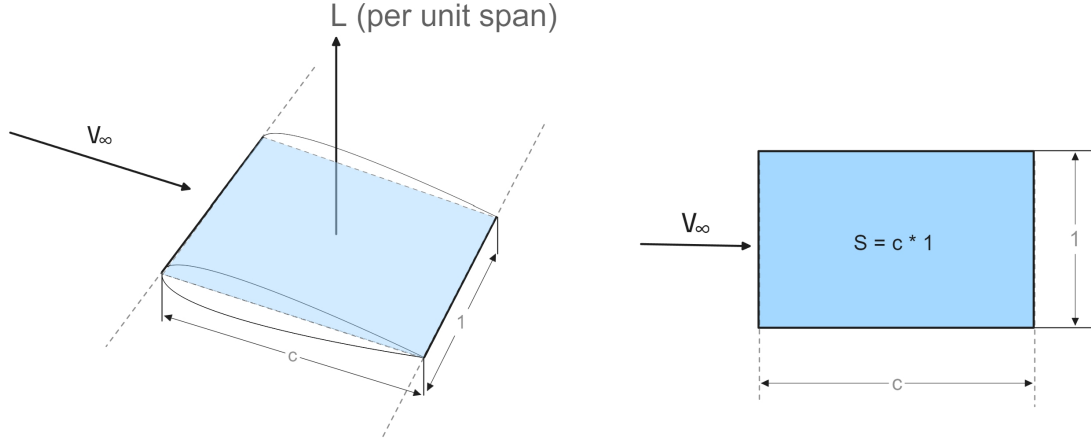


Figure 2.8: Illustration of unit span l and unit area S

$$\begin{aligned}
 L &= \frac{1}{2} \rho V_{\infty}^2 S C_L \\
 D &= \frac{1}{2} \rho V_{\infty}^2 S C_D \\
 M &= \frac{1}{2} \rho V_{\infty}^2 S_{MGC} C_M
 \end{aligned} \tag{2.19}$$

Where S_{MGC} is the reference wing area with chord equal to the mean geometric chord (MGC), C_L the 3D lift coefficient, C_D is the 3D drag coefficient and C_M is the 3D moment coefficient. The ability to calculate lift, drag and pitch moment based on pre-defined factors such as AoA is critical for control allocation in subsequent phases. In this process, control inputs are distributed to different actuators to achieve desired responses from the airship. Consequently, modelling the non-dimensional coefficients C_L , C_D and C_M becomes essential to determine the lift, drag and pitch moment generated at given parameters.

In order to understand how lift and drag vary with changes in AoA α , this relationship is often graphically represented by what is known as a lift and drag curve, or commonly known as a polar. The lift polar illustrates the relationship between the lift generated by an airfoil or wing and the AoA at which it operates, and makes the change visually comprehensible. The lift is usually represented by the lift coefficient and the same is true for the drag curve. In Figure 1.9 is an example of lift and drag curves. At low AoA the lift

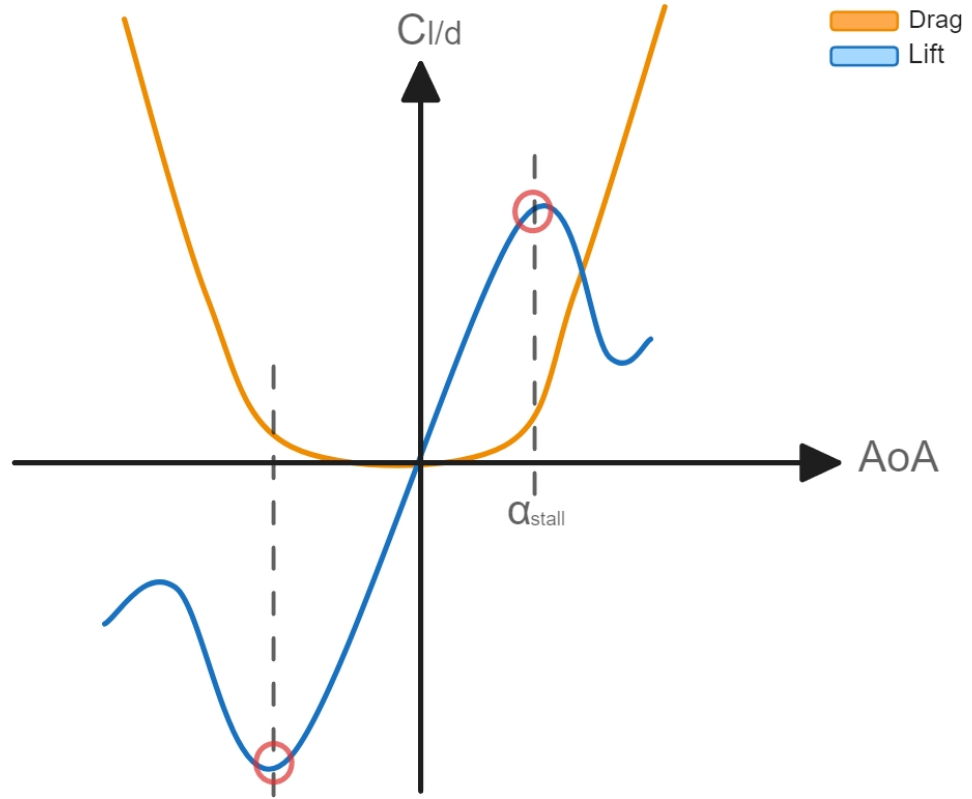


Figure 2.9: Common shapes of lift and drag curves with with clearly identifiable stall angle α_{stall})

curve has a linear relationship, with lift increasing in proportion to AoA. This region is often referred to as the linear or pre-stall region, where the airflow over the wing remains attached and the lift coefficient increases steadily. As the AoA continues to increase, the lift curve reaches a point where the airflow over the wing becomes separated, leading to a sudden decrease in lift and a sharp increase in drag. This phenomenon is known as stall which denotes the limit of the wing's operating envelope and the AoA at this stage is called α_{stall} [9, Chapter 8] [14].

Another property related to the lift curve is the lift curve slope Cl_α . It describes how efficiently an airfoil generates lift as its AoA changes, and mathematically it is the derivative of lift L with respect to AoA α [16]:

$$C_{l_\alpha} = \frac{dC_l}{d\alpha} \quad (2.20)$$

A larger lift curve indicates a more efficient lift generation of the airfoil over a range of AoA.

2.2.7 Trailing Edge Flap

The trailing edge flap is a major aerodynamic component located along the trailing edge of the wing. It is designed to modify the aerodynamic properties of the wing by altering its lift and drag characteristics by deflecting the flap up or down [9].

If the flap is deflected downwards, it increases the camber of the airfoil and shifts the lift curve upwards, but the linear range becomes narrower as α_{stall} gets closer to the origin which means stall arrives earlier; if the flap is deflected downwards, the trailing edge flap has the opposite effect.

Among the many trailing edge flap designs used in aviation, notable variants include the plain flap, split flap, flowler flap, slouted flap, double slouted flap and zap flap. Various kinds of trailing edge flap are depicted in figure 2.10. Each of these flap configurations has its own advantages and limitations. The wing of the airship Looncraft Squid is deflected by a plain flap, also known as a simple flap or hinge flap, which is one of the most basic types of trailing edge flap used on conventional aircraft. It consists of a single hinged segment that rotates about its hinge point to extend downward when deployed and retract into the trailing edge of the wing when stowed. Unlike slotted flaps, the plain flap does not have a wider slot to allow the high pressure air to flow through the slot to the other side; the lift and drag performance may be less outstanding compared to other flapping designs, but it is simple both in terms of construction and operation. In Looncraft Squid, the hinge point is on the chord line, so flapping up results in the wing being mirrored downwards at the same angle. [4] [5] [14]

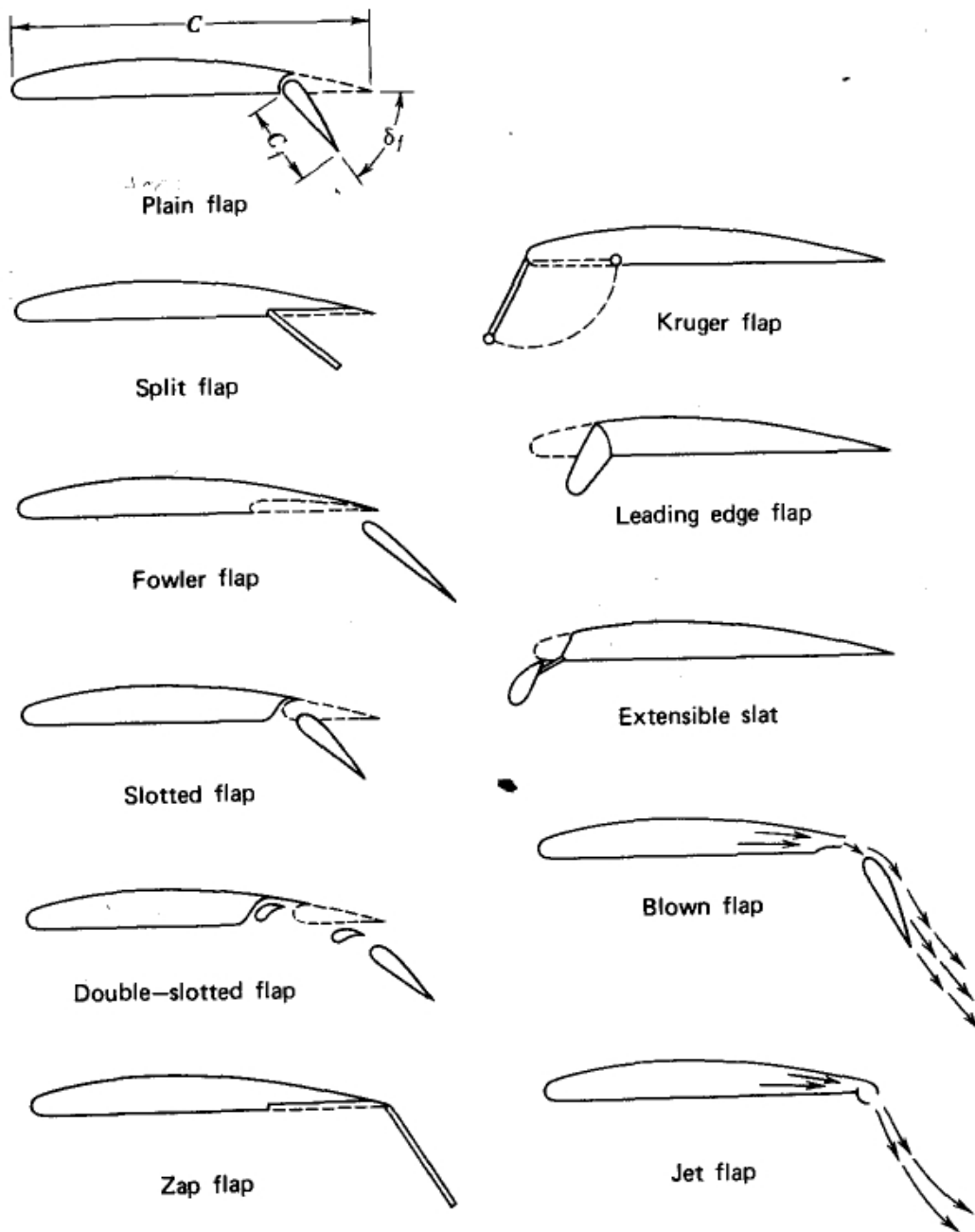


Figure 2.10: Various types of trailing edge flap. [14]

3 Methodology and Approach of nondimensional coefficient modelling

In this chapter, methodologies and approaches employed in modeling of nondimensional coefficients are discussed. The accurate determination of the lift-, drag-, and moment coefficients is crucial for understanding the aerodynamic performance and characteristics of airfoils, which in turn influences the design and optimization of the fin's control allocator. To provide a comprehensive analysis, two primary approaches are tried: the analytical approach and the numerical approach.

3.1 Analytical approach

A model of the nondimensional coefficients C_l , C_d and C_m is required to calculate the actual dynamic forces and moments, which are essential for subsequent control allocation. These calculations take place in a real time environment, so the acquisition of memory resources and the computational effort should also be taken into account.

Anderson et al. [4] provides an useful approach to modell which shows that the lift, drag and moment coefficients are each a function of AoA α , Mach number M_∞ and Reynolds Number Re :

$$\begin{aligned}C_l &= f(\alpha, M_\infty, Re) \\C_d &= f(\alpha, M_\infty, Re) \\C_m &= f(\alpha, M_\infty, Re)\end{aligned}\tag{3.1}$$

Due to the low speed of the airship during the flight, which is significantly lower than 0.3 times the speed of sound ($M_\infty \ll 0.3$), the Mach number M_∞ can be disregarded. The

3.1 Analytical approach

reason for this will be explained later, the coefficients can now only depend on AoA and Re:

$$\begin{aligned} C_l &= f(\alpha, Re) \\ C_d &= f(\alpha, Re) \\ C_m &= f(\alpha, Re) \end{aligned} \tag{3.2}$$

Anderson et al. [4] first introduces the pressure coefficient C_p , which is a dimensionless quantity describing the pressure distribution over the upper surface around the airfoil. The pressure coefficient is defined as [4, equation. 5.23]

$$C_p = \frac{p - P_\infty}{q_\infty} \tag{3.3}$$

where:

- P the local static pressure at a certain point on the surface;
- P_∞ the freestream static pressure;
- q_∞ the dynamic pressure of the flow, given by $q = \frac{1}{2} \rho V_\infty^2$;

Figure 3.1 shows an example of the pressure coefficient over the top and bottom surfaces, over the entire chord length.

The lift coefficient can be calculated by integrating the C_p differences between the top and bottom surfaces of the airfoil over the chord length and then dividing by the chord length again:

$$C_{l_0} = \frac{1}{c} \int_0^c C_{p_l} - C_{p_u} dx \tag{3.4}$$

with:

- C_{p_l} pressure coefficient on lower surface
- C_{p_u} pressure coefficient on upper surface

The lift coefficient needs a compressibility correction if M_∞ is infinite at subsonic speed:

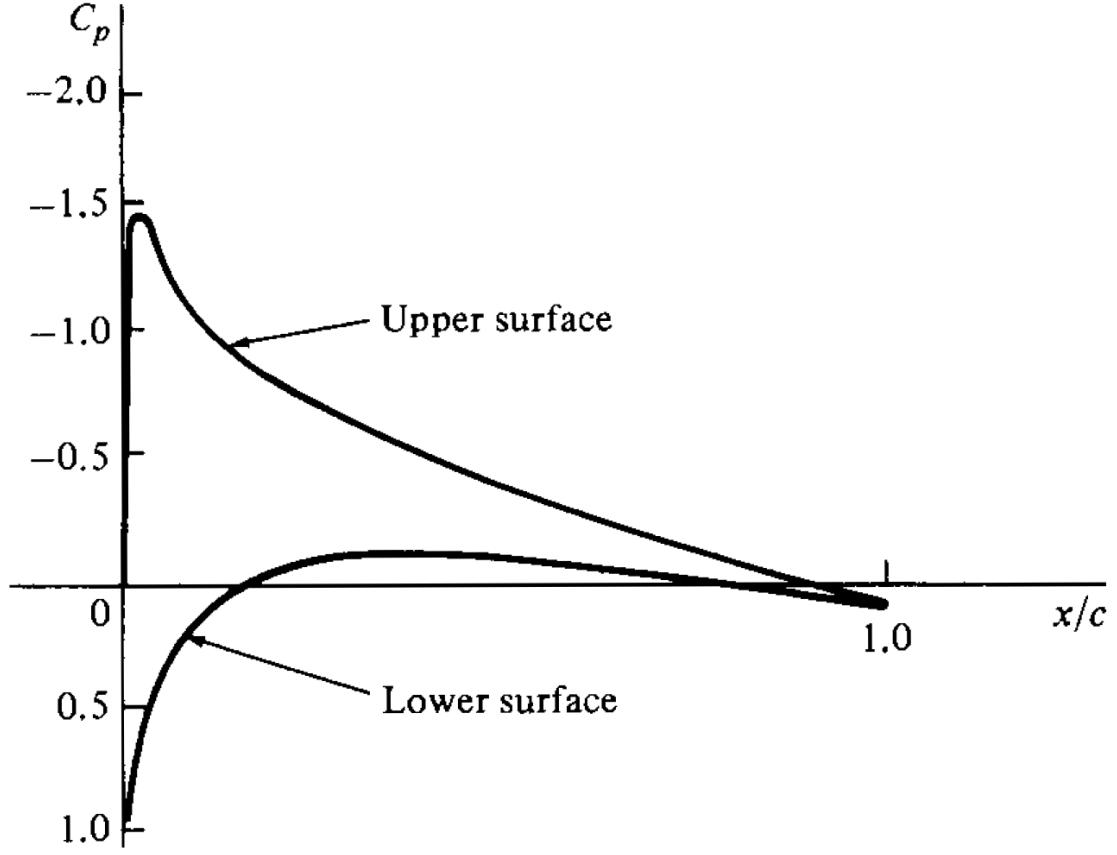


Figure 3.1: Distribution of the pressure coefficient C_p each of upper and lower surface, using the example of NACA0012. [4, figure 5.11]

$$C_l = \frac{C_{l_0}}{\sqrt{1 - M_\infty^2}} \quad (3.5)$$

However, due to low speed of the airship during the flight ($M_\infty \ll 0.3$), the denominator of the equation 3.5 is nearly equal 1, hence the Mach number can be neglected:

$$C_l = C_{l_0} = \frac{1}{c} \int_0^c C_{p_l} - C_{p_u} dx \quad (3.6)$$

Similarly, the total drag's coefficient C_D is composed of profile drag coefficient C_d and induced drag coefficient C_{D_i} where the profile drag C_d is obtained by the sum of skin

3.1 Analytical approach

friction drag C_{df} and pressure drag coefficient and C_{dp} , and the induced drag is obtained via [4, equation 5.47]

$$C_D = C_{d,} + C_{D,i} \quad (3.7)$$

where

and AR the aspect ratio between wingspan b and the wing's surface S : [4, equation 5.45]

$$AR = \frac{b^2}{S} \quad (3.8)$$

Anderson et al. also provides an approach for calculating the pitch moment coefficient C_M . It consists of a sum of the pitch moment where the deflection is zero (C_{m_0}) and another part caused by the AoA and the deflections:

$$C_{M,cg} = C_{m,0} + \frac{dC_m}{d\alpha}\alpha + \Delta C_M \quad (3.9)$$

with

$$\Delta C_M = -V_H \frac{dC_L}{d\delta}\delta + \Delta C_M \quad (3.10)$$

V_H is a specific value given by the airplane, hence this approach is hardly practical in this proposal.

The approach of Anderson et al. is a parametric solution that makes the lift, drag and pitch moment coefficients scalable. In this approach, the lift coefficient serves as the basis for calculating the drag and pitch moment coefficients. Despite the simplification of the lift coefficient due to the low speed condition where the Mach number and compressibility correction are neglected, there are still some unknown terms that require further investigation. The first problem is the free stream static pressure p_∞ . This value is obtained by measurements in the wind tunnel using pitot tubes, pressure transducers or pressure taps at different locations in the flow field. Another solution to obtain p_∞ is to use CFD simulations based on the mathematical models of fluid flow. However, both methods are challenging. The difficulty is transferred to the drag and pitch moment coefficients, since both coefficients require a given C_l value in the calculation. To calculate the pitch moment coefficient, the slope $\frac{dC_L}{d\delta}$ and the slope $\frac{dC_m}{d\alpha}$ are required, and the V_H of the airship is unknown in this case.

Li et al. [12] provides another method for modelling the lift, drag and pitch moment coefficients. It is a parametric method specifically designed to estimate the incremental change in lift, drag and pitch moment coefficients ΔC_l , ΔC_d and ΔC_m due to the control

3 Methodology and Approach of nondimensional coefficient modelling

surface deflection δ . Li et al. point out the observation that the inclusion of fins results in an additional normal force acting on the hull of the airship. The lift coefficient increment can be obtained as [12, equation 24]:

$$\Delta C_l = C_{L\alpha} \cdot \tau \cdot \eta_d \cdot k_{3D} \cdot \delta \quad (3.11)$$

where:

$$\begin{aligned} \tau &= 1 - \frac{\theta_f - \sin(\theta_f)}{\pi} \\ \theta_f &= \cos^{-1}\left(\frac{c_f}{c-1}\right)k_{3D} \end{aligned} \quad (3.12)$$

with:

c_f	flap chord
c	airfoil chord
η_d	correction factor derived from experimental data and expressed as a function of the basic deflection
k_{3D}	efficiency factor related to tau that considers 3-D effects.

Li et al. provide curves for η_d and k_{3D} for the case where $\tau=0.5$ in line graphs 3.3. The increment of drag coefficient is provided as follows [12, equation 26]:

$$\Delta C_d = 1.7 \left(\frac{c_f}{c}\right)^{1.38} \frac{S_f}{S_{FA}} \sin^2(\delta) \quad (3.13)$$

where:

S_f	flap's surface
S_{FA}	fin's surface

The estimation of how flap deflection impacts on the pitching moment coefficient can be derived using thin airfoil theory, which is given by the ratio of increment of pitching moment at one quarter of chord length $\Delta C_{m_{\frac{1}{4}}}$ to increment in lift ΔC_L [12, equation 27]:

$$\frac{\Delta C_{m_{\frac{1}{4}}}}{\Delta C_L} = -\frac{2\sin\theta_f - \sin 2\theta_f}{8(\pi - \theta_f + \sin\theta_f)} \quad (3.14)$$

The parametric method used by Li et al. to calculate the coefficient increments allows a quantitative assessment of the relationship between the coefficients and the deflection

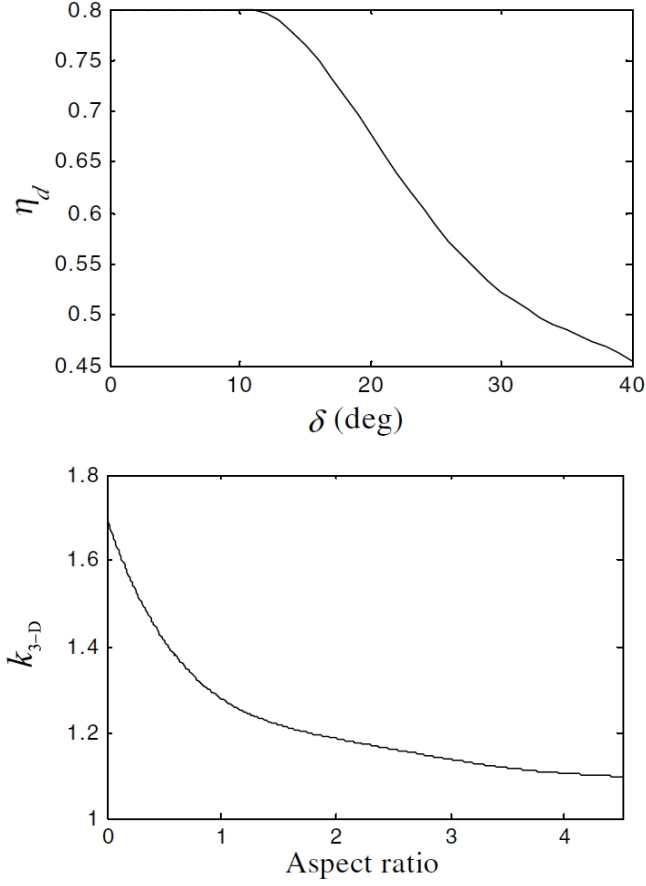


Figure 3.2: η_d for plain flap, [12, Fig. 10]

Figure 3.3: 3-D effectiveness factor by $\tau = 0.5$

of the control surface. However, from the line graph of η/δ values it is not possible to read the values for $\delta \leq 10$; also the flap-chord ratio c_f/c by the value of $1/3$ gives a τ value by 0.69 rather than 0.5, therefore the values from the line graph of both η_D and 3D effectiveness factor are not plausible to use in this proposal, especially for calculating the lift coefficient increment ΔC_l and ΔC_m .

With regard to the subject under discussion, the coefficients C_l , C_d and C_m are affected by the control surface deflection delta as well as the AoA α . Hasan [11] points out that the total lift coefficient should be made up of a linear combination of the baseline airfoil value where the deflection is zero C_{l_0} and a coefficient increment term $\Delta C_l(\delta)$ caused by the deflection:

3 Methodology and Approach of nondimensional coefficient modelling

$$C_l = C_{l_\alpha}(\alpha - \alpha_0 + k\delta) = C_{l_0} + \Delta C_l(\delta) \quad (3.15)$$

with k being a function of the flap chord ratio E :

$$k = \frac{1}{\pi} \left\{ \cos^{-1}(1 - 2E) + 2\sqrt{E(1 - 2E)} \right\} \quad (3.16)$$

This equation also shows that the AoA α and the deflection angle δ can be linearly combined, so that the lift coefficient C_l can be calculated from the lift curve slope C_{l_α} using an angle composed of the current AoA α and the deflection angle δ ($\alpha_0 = 0$ when the wing is horizontal to the fuselage).

In Hasan's approach, the slope C_{l_α} is obtained by ARC2D, a computational fluid dynamics (CFD) program developed at NASA Ames Research Center. The program utilizes finite-difference techniques to solve two-dimensional Euler equations and Navier-Stokes equation and is specifically designed for analyzing two-dimensional aerodynamic problems, particularly focusing on airfoils and similar geometries that are simply connected. [15]

Like the sectional lift coefficient of the airfoil, the drag coefficient can be made up of a base term without deflection and another term due to deflection:

$$C_d = C_d(\delta = 0) + C_{d_\delta} \quad (3.17)$$

Where C_{d_δ} the increment due to the deflection and is approximated by a best second- and third- order polynomial fits:

$$C_{d_\delta} = a_1\delta + a_2\delta^2 + O(\delta^3, \delta^4) \quad (3.18)$$

The airfoil's quarter chord moment coefficient C_m is given by:

$$C_m = C_{m_0} - m \cdot \delta \quad (3.19)$$

where:

$$m = \frac{\alpha_0}{\pi}(1 - E)\sqrt{E(1 - E)} \quad (3.20)$$

and $C_{m_0} = 0$ for NACA0015 [11].

Hasan goes further in his approach and approximates the total drag coefficient C_d using the best fifth- and sixth-order polynomial fits to the values obtained by the ARC2D programme. The values of the polynomial coefficients given are not directly applicable, since Hasan's model is mainly for the NACA0015 airfoil, but with a different chord to flap ratio E , and the values given in the appendix are mainly aimed at $Re = 3$ million, which is already on the edge of consideration in this proposal. Nevertheless, Hasan provides a valuable perspective by utilising CFD for simulation, followed by polynomial modelling of the simulation results.

3.2 Numerical Approach

Following by the perspective mentioned above, utilising CFD for simulation has emerged as another approach to modeling airfoil characteristics. This section delves into the application of CFD for simulating the lift, drag, and moment coefficients of airfoils. Unlike the analytical methods, which rely on theoretical simplifications and assumptions, CFD leverages computational power to solve the complex equations governing fluid flow around airfoil geometries. This section describes in detail the work and the approach, starting with the modelling of the non-dimensional coefficients with Xflr5, followed by the implementation of the Gazebo simulation environment and rudder's control allocation.

3.2.1 Introduction of Xflr5

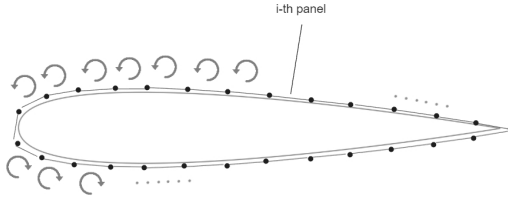
The Xflr5 is an airfoil design and analysis software package. With Xflr5, users can analyse the aerodynamic characteristics of airfoils, including lift, drag and pitching moment coefficients, over a range of angles of attack and Reynolds Number. The Xflr5 also allows users to predict the performance of aircraft configurations, including lift-to-drag ratios, stall behaviour and trim conditions such as control surface deflection. It provides visualisation tools to plot airfoil polars, pressure distributions and velocity contours. Another feature is wing design: users can design and analyse wings and wing sections with Xflr5. The software provides features for designing wing geometries, including planform shape, twist and dihedral angle. It also provides tools for predicting wing performance, including lift, drag and stability characteristics. Xflr5 is freely available to download and use, giving users access to the latest features, but does not guarantee the accuracy of the results. The latest version is Xflr5 V6, which is used in this proposal. [17]

Given that the chosen airfoil is NACA0015, the primary focus of this proposal with Xflr5 is the airfoil analysis through a range of angles of attack and control surface deflections. This is facilitated by the inclusion of the Xfoil program in Xflr5. The Xfoil program

3 Methodology and Approach of nondimensional coefficient modelling

allows users to utilise the capabilities of Xfoil, but also to access additional tools and features for wing and aircraft analysis within the same software environment. Xfoil uses the panel method to calculate the aerodynamic forces and moments acting on an airfoil. This is an approximation method in which the airfoil geometry is discretised into a series of small panels: in 2D panel methods, each panel represents a straight line segment along the contour of the airfoil, and in 3D panel methods, each panel represents a small surface area on the airfoil surface where all four corner points are in a plane.

2D Panel Method:



3D Panel Method:

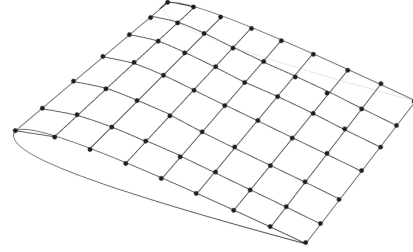


Figure 3.4: 2D- and 3D-Panel Method

In higher-order panel methods, the geometry of the body is represented by a series of higher-order line curves, such as B-spline curves, which provide greater flexibility and accuracy in capturing the details of the surface geometry and allow a more accurate description of the flow field near the wing surface. For each panel, flow velocities and pressures are calculated at discrete points and summed up together [17]. Due to the accessibility of the software, xflr5 is used for airfoil analyse in this proposal and is the main source of the non-dimensional coefficients C_l and C_d .

3.2.2 Non-Dimensional Coefficient Estimation using xflr5

As introduced in the previous chapter, the Looncraft Squid is equipped with 4 fins, each with a deflectable rudder (control surface). The purpose is to take advantage of the aerodynamic forces and reduce the usage of side thrusters for better energy efficiency and a longer flight duration. In order for the system to determine an appropriate command $u_{act, d_{fin}}$ for rudder's deflection control, it requires knowledge about how the rudder's control surface affects the airships dynamic. The workaround is to create a parameter list for calculating the force's non-dimensional parameter with given flight conditions and deflections.

3.2 Numerical Approach

The airfoil analysis using Xfoil starts by defining the shape of the airfoil under "Direct Foil Design". The airfoil "NACA0015" is predefined in the programme under "NACA Foils", in this case it is not necessary to redraw the waypoints of the contour manually, nor is it necessary to enter the percentage of thickness and camber. The contour model consists of 150 waypoints, the number of points varies by 1 to 3 points for airfoils with later deflection. The predefined shape fulfils the requirement of an airfoil in neutral position (horizontal with $\alpha=0$) without deflection. It is named "NACA 0015 Flap neutral". In order to model airfoils with deflections, further contour shapes are defined in parallel, whose TE Flap (Trailing Edge) is stored with the respective degree of deflection. The deflection range is discretised in 5° increments, resulting in 18 additional contour shapes with each being saved as a stand along airfoil profile. The chord length of the Looncraft Squid's rudder is one third of the chord length; the measurement starts from the Trailing edge, so the hinge of the deflectable rudder is at the position at 66% of the chord length and 50% for the vertical hinge position which means vertically in the middle. The profiles are shown in the figure.

The fins of the airship have a trapezoidal shape. For the purposes of simplicity, a rectangle is considered for the simulation instead, the width of which corresponds to an average value of chord lengths at the root and wingtip and is constant. For the simulation, the average between the root's and tip's chord length is used to for the airfoil analysis. The objective is to identify the non-dimensional coefficients of the fins at different air speeds. It should be noted that the coefficient values C_l and C_d are dependent on the AoA and rudder deflection. Consequently, the coefficients are contingent upon three variables: AoA, air speed and deflection. The Looncraft Squid is omni-directional, so the intended operating range of AoA is from -90 to 90 degrees. The increment for AoA is set to 0.5 in order to achieve a compromise between the programme's calculation time and accuracy, otherwise the Xflr5 programme may crash before the analysis comes to the end. The rudders of the 4 fins can each deflect by 45° upwards and downwards, with deflecting upwards defined as positive. The grip points for the deflection are listed in table 3.1. The maximum relative air speed during the flight is currently intended to be below 30 km/h. This specification can be improved in the future development, therefore a maximum air speed of up to 60 km/h is assumed for the task of searching for non-dimensional coefficients in order to ensure sufficient redundancy. For simplicity, each of the airspeed listed in the table 3.1 is referred to as a velocity class in this proposal, or class for short.

3 Methodology and Approach of nondimensional coefficient modelling

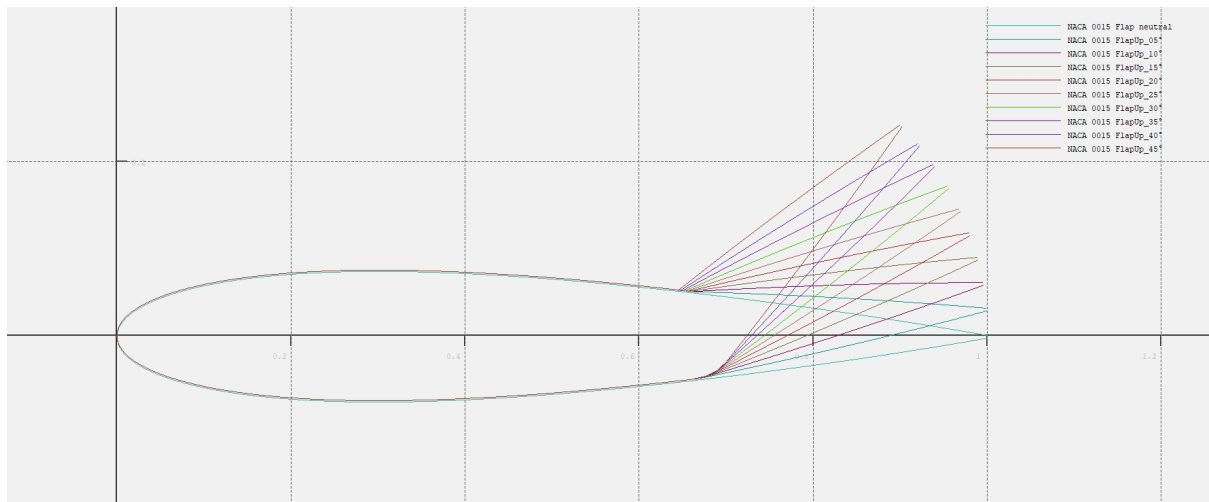


Figure 3.5: Deflected airfoil shapes creating in Xflr5's "Direct Airfoil Design tool".

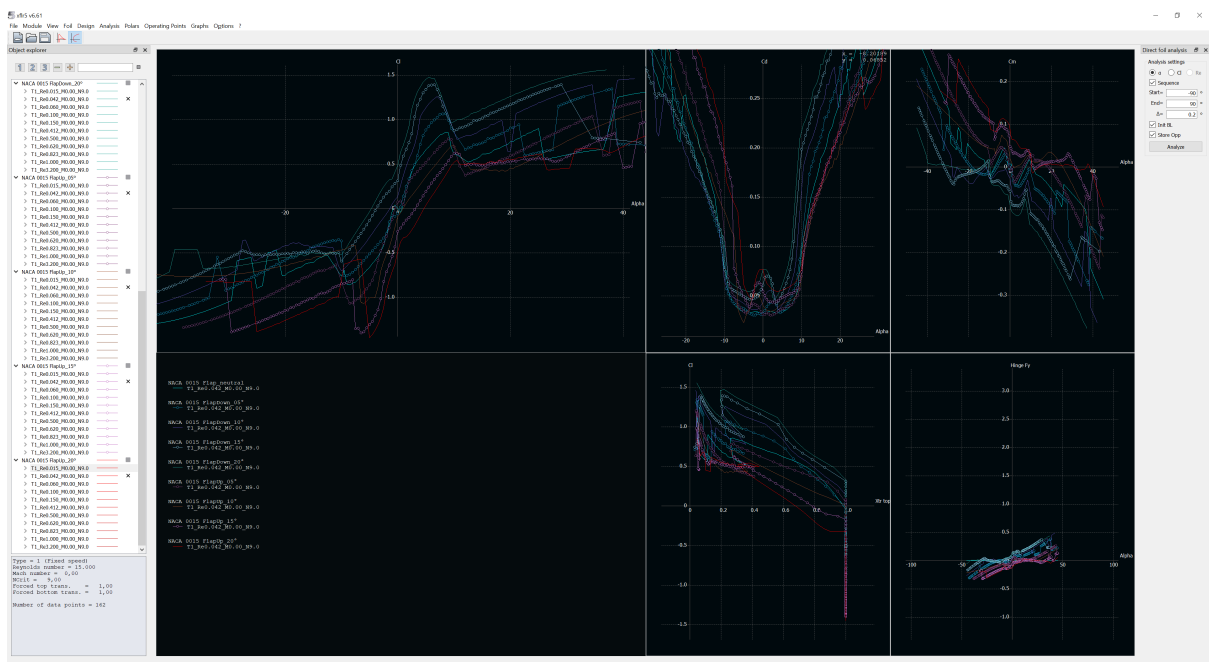


Figure 3.6: Airfoil analysis in Xflr5

3.2 Numerical Approach

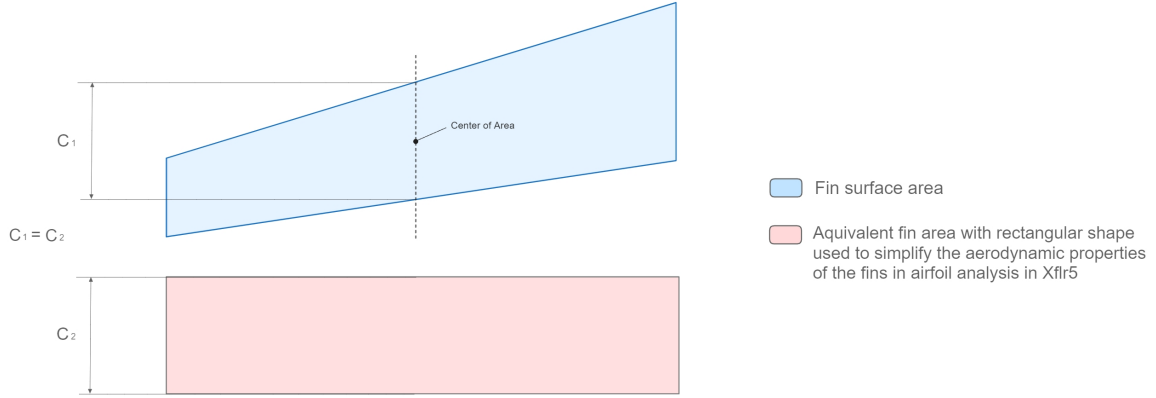


Figure 3.7: Fin area shape comparison

δ [°]	0	5	10	15	20	25	30	40	45
δ [°]	0	-5	-10	-15	-20	-25	-30	-40	-45

Table 3.1: Considered deflection grip points in -45° to 45°

V_∞ [km/h]	5	10	15	20	25	30
Re [-]	31286	62573	93859	125146	156433	187719
V_∞ [km/h]	35	40	45	50	55	60
Re [-]	219006	250293	281579	312866	344153	375439

Table 3.2: Considered velocity and Reynolds Number

The XFOIL programme provides the option of conducting 'batch analyses', which requires predefined airfoil shapes, Renold's number (which is depending on the air speed) and AoA for nummerical simulation. In XFOIL, user is allowed to specify up to a maximum of 12 Reynolds Numbers in one batch analysis, which are subsequently incorporated into the nummerical simulation. To ensure a linear observation, the range of air speed from 0 to 60 km/h is divided by 12, resulting in an equal moderate distance of 5 km/h between the control points. The airspeed and the corresponding Reynolds Number are listed in the table 3.2. The numerical results of $C_l(\alpha)$ and $C_d(\alpha)$ are shown in figure 3.8 and 3.9. For the sake of simplicity, the curve of $C_l(\alpha)$ and $C_d(\alpha)$ will be referred to respectively as the C_l - and C_d -curve.

3 Methodology and Approach of nondimensional coefficient modelling

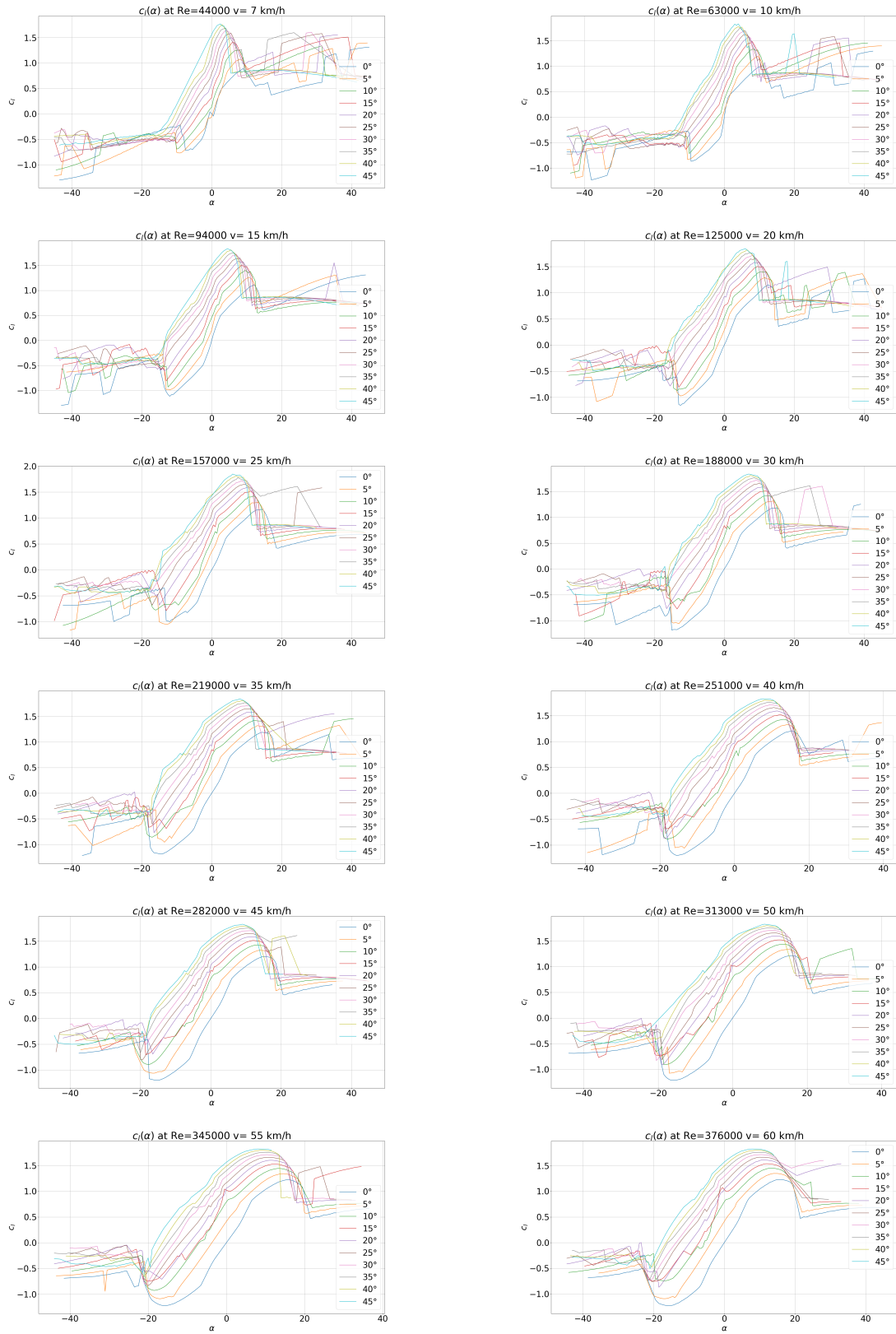


Figure 3.8: curve family of $C_l(\alpha)$, each family represent one of the airspeed and Reynolds Number listed in table 3.2.2

3.2 Numerical Approach

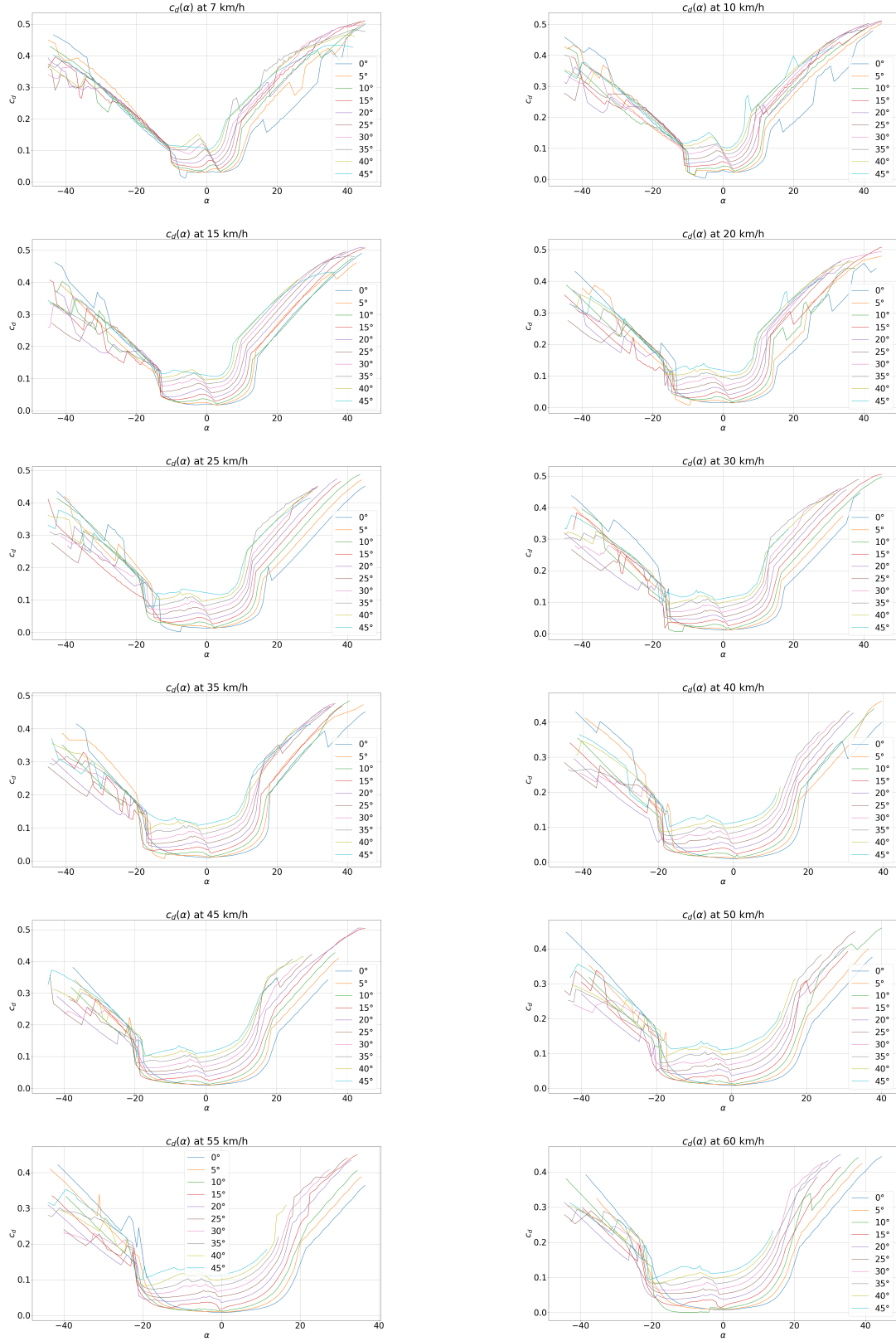


Figure 3.9: curve family of $C_l(\alpha)$, each family represent one of the airspeed and Reynolds Number listed in table 3.2.2

Each curve family corresponds to a velocity class (each velocity corresponds to a Reynolds Number) and includes 10 shape-like curves, slightly offset from each other, with each curve corresponding to a deflection value. One of the most important observations is that the curves with negative deflection are mirror-symmetrical to the curves with positive deflection. For the sake of simplicity, only curves of deflection from 0° to 45° will be discussed. A more comprehensive analysis is following in the next chapter.

3.3 Development of a Parameter List for Control Allocation

This section deals with the design and creation of the parameter list. The parameter list is applied in a real-time environment, as it runs on the flight controller Cube Orange and is repeatedly queried during runtime for control assignments. This requires a reduced complexity of the parameter list and the handling with it.

The initial approach is to include all function points into the LookUpTable and apply interpolation for values in between. It comes up with millions of values for the range of AoA, deflection and velocity defined above, and storing this raw function points costs lots of memory. The memory requirement can be reduced by reducing the resolution and moving the included function points further apart on the diagram so that fewer function points are included in the look-up table. When looking up for a value, it still requires a lot of iteration until finding the appropriate gap to do the interpolation. Another approaches that result in a smaller package size and less time consuming to required. Two approaches are under consideration.

3.3.1 Neglecting of Pitch Moment

When analysing airfoils and wings, the pitch moment coefficient C_m , acts around the fin's Y-axis and affects the inclination or tilting of the wing. With regard to the significance of pitch moment, it should be noted that the pitch moment generated by lift is far more relevant than the pitch moment around the fin's Y-axis discussed here. Since the lift coefficient is fully greater than the pitch moment coefficient and the distance from CoV to the respective fins is significantly greater than 1 metre, the product of both lift coefficient and distance from CoV to the respective fins is even more significantly greater than the pitch moment coefficient around the fin's Y-axis. Neglecting this moment leads to simplification of the calculations and model without significantly affecting the accuracy and reliability of the aerodynamic analysis. There is also another reason which supports the neglecting of C_m . In X-configuration, each fin has an opposite fin that generates a

pitch moments in reverse clockwise direction so that the pitch moments mostly cancel each other out in pairs. The small unbalancing of this pitch moment can be absorbed by the elasticity of the airship's hull. For the reasons mentioned, the coefficient is neglected in the rest of the work.

3.3.2 Approximation of C_l and C_d

The first approach is to directly approximate the curves of C_l and C_d using polynomials. The general shape of the C_l and C_d curves has been introduced in figure ?? . A C_l -curve in the laminar region is approximately linear to α , so it makes sense to write an C_l -curve as an approximately linear function of α . A C_l -curve can also be approximated by a polynomial of higher degree, in particular by an odd function such as a 3rd degree polynomial, in order to better represent the contour when getting closer to stall angle α_{stall} . However, this leads to increased complexity when solving, as functions of higher degree could have one, several or even no solutions, as long as the degree of the function is higher than one. This requires algorithms for case differentiation and error handling if no solution is found.

A quadratic function is used to approximate a C_d -curve due to the symmetry to the Y-axis. An even function of higher degree is not more effective, as the C_d -curve no longer has a strictly concave shape up to a certain point, instead, the slope of the curve slows down and the curve becomes a little flatter. For this reason, the quadratic function remains the best choice for the approximation, which is convincing both in terms of shape and simplicity, especially for the laminar region.

3.3.3 Approximation of C_n and C_a

As presented in Figure ?? , a fin-fixed coordinate system is introduced exclusively for each fin. In this approach, two forces are considered: normal force F_n , which is always normal to the fin's airfoil chord and points to the negative z -direction, and the axial force F_a parallel to the airfoil's chord which acts in the negative x -direction. Similar to lift and drag, the behaviour of the both normal and axial forces can be related by their non-dimensional coefficient C_n and C_a as in (2.19):

$$\begin{aligned} F_n &= \frac{1}{2} \rho V_\infty^2 S C_n \\ F_a &= \frac{1}{2} \rho V_\infty^2 S C_a \end{aligned} \tag{3.21}$$

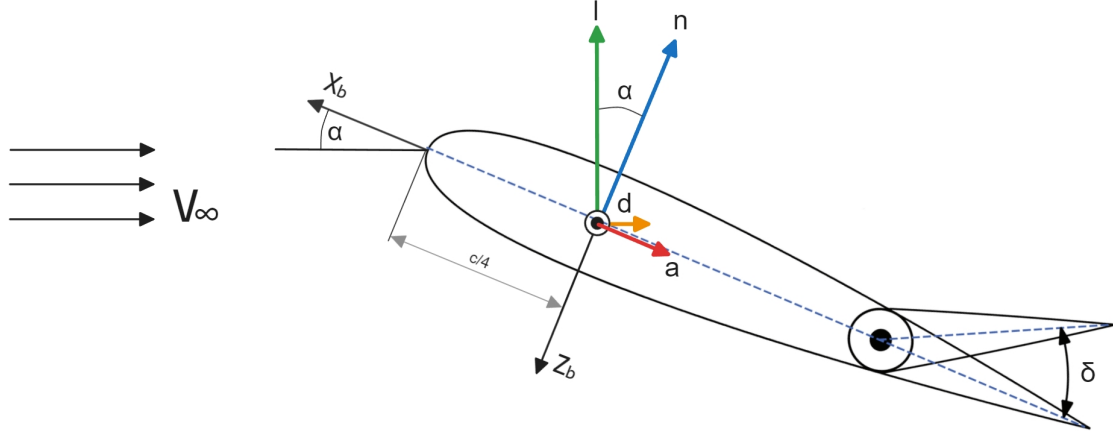


Figure 3.10: Simplification

Figure 3.10 shows the relation between CnCa and ClCd. CnCa is acquired by applying a rotation matrix around AoA α :

$$\begin{bmatrix} C_n \\ C_a \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} C_l \\ C_d \end{bmatrix} \quad (3.22)$$

It is assumed that the axial force C_a is considerably smaller and negligible compared to C_n . Consequently, only the non-dimensional coefficient C_n is modelled in the CnCa-approach. The following chapter will undertake a detailed examination of this assumption and present evidence in support of it. This assumption confers a significant advantage to the approach, as it eliminates a variable and substantially diminishes the subsequent control allocation effort. Two methods exist for modeling C_n . The C_n and C_a values can be obtained from the approximated C_l and C_d curves, converted through rotation (3.22). In this method, a coordinate transformation needs to be performed each time, and it always requires pairwise related C_l and C_d . Another way is to prepare a parameter list for C_n by fitting the C_n curve directly. In this process, the function points from C_l and C_d are transferred pairwise via the coordinate transformation in C_n and C_a , as illustrated in the aforementioned equation.

3.3 Development of a Parameter List for Control Allocation

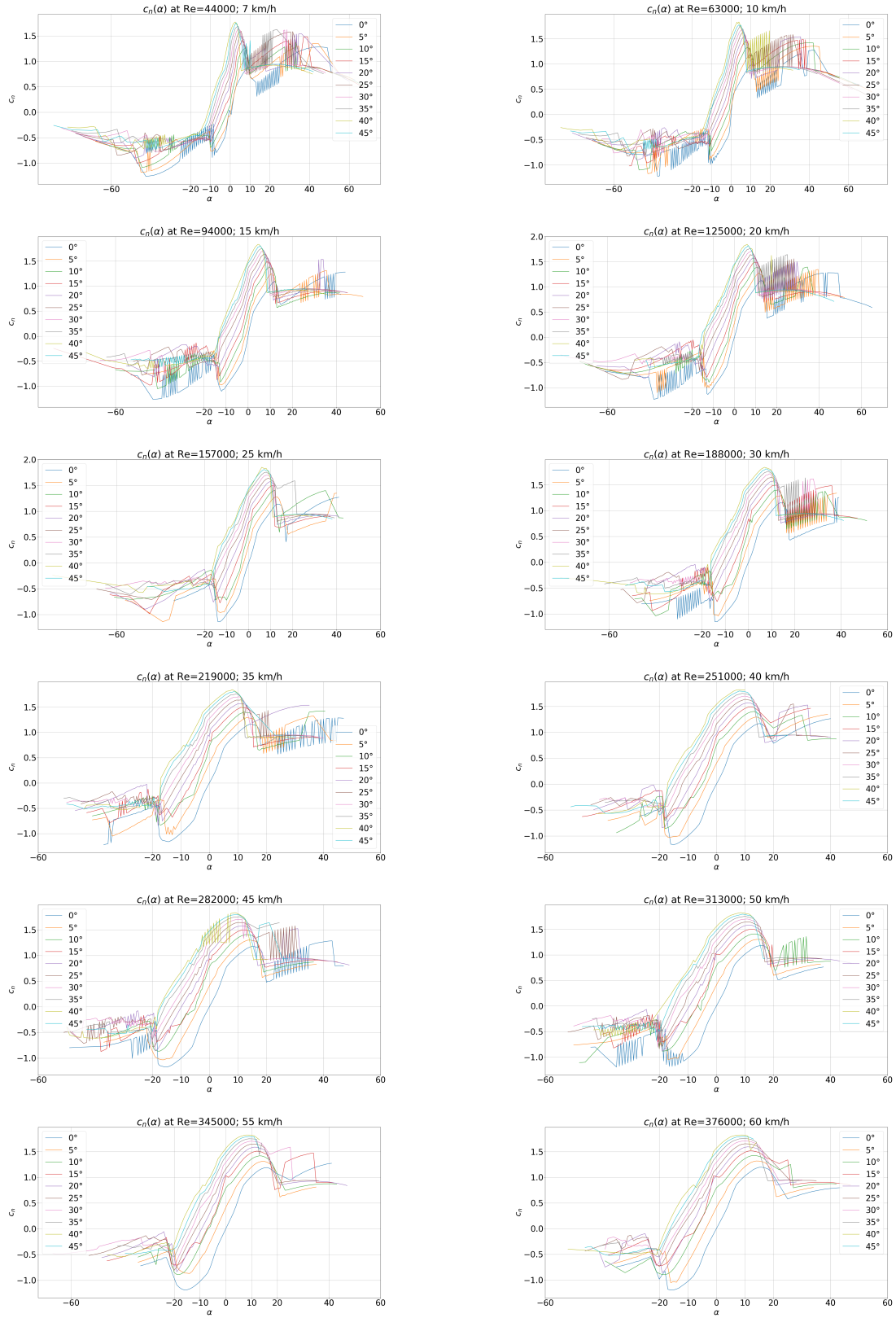


Figure 3.11: curve family of $C_n(\alpha)$, each family represent one of the airspeed and Reynolds Number listed in table 3.2.2

3 Methodology and Approach of nondimensional coefficient modelling

The C_n curve is approximated by a linear function. As like the modelling approach for C_l curves, the C_n -curves have a similar similar shape and is able to be approximated with a linear straight line either. Therefore, the function $C_n(\alpha)$ can be expressed as a linear line:

$$C_n(\alpha) = m \cdot \alpha + b \quad (3.23)$$

For any given α , the value for C_n can be calculated, as long as the AoA doesn't come over the α_{stall} . In comparison to the process of hardcoding, the approximation of the curve brings in a significant reduction of complexity, as only the coefficient a and b of each curve points are required.

3.3.4 Parameterisation of Deflection

After approximating the relationship between C_n and α by a linear line function, the focus moves on to managing the displacement of curves parameterized depending on the deflection. As shown in figure 3.11, the curves of a velocity class have a relatively similar shape and appear with different upward displacement depending on the deflection rudder's deflection. A regularity is noticable in the diagram, especially when looking at the heights and depths at those two stall angles ($\alpha = \alpha_{stall}$), the course of the displacement is almost straight, but the distances between the curves become smaller and smaller. For each set of curves, only the neutral curve (the curve corresponding to zero deflection) is modeled as the baseline using equation (3.23). As the curves are parallel shifts from baseline, the curve's slopes are very similar, so the coefficient a (which represents the baseline's slope) is approximated with a linear line; as the distances between the curves become smaller and smaller as the displacement increases, an attempt is made here to approximate the offset of the baseline (which corresponds to the coefficient b) with a quadratic polynomial. A higher-degree polynomial can lead to several solutions when solving for δ , which increases the algorithmic complexity. The choice of degree is evaluated in the next chapter. This approach completes the equation (3.23) to the following form:

$$\begin{aligned} C_n(\alpha, \delta) &= m \cdot \alpha + b \\ m &= m_1 \cdot \delta + m_0 \\ b &= b_2 \cdot \delta^2 + b_1 \cdot \delta + b_0 \end{aligned} \quad (3.24)$$

3.3.5 Stall Boundries

The magnitude of C_n drops in when AoA is beyond α_{stall} and modelling the curve outside the laminar range becomes difficult. For this reason, only the laminar range is taken into consideration and modeled, and the curve is cut off at the amplitudes (where $\alpha = \alpha_{stall}$) and the range within is the entire area where the deflectable rudder can choose its operating point. A set point beyond this cannot be met and needs to be forwarded to the control allocator of the side thrusters. From the algorithm's point of view, there is a demand to know whether the rudder's achievable operating point (due to the current AoA) is able to cover the set point's demand. For this purpose, it is necessary to include the boundaries into the lookup table, which results 2 curves: the upper- and lower boundary. Since the displacement of the curve's amplitudes nearly resembles a straight line, it is attempted to approximate the displacement of the amplitudes with a straight line which results 2 coefficients per boundary curve.

$$\begin{aligned} C_{n,upperbound}(\alpha) &= m_{upper} \cdot \alpha + b_{upper} \\ C_{n,lowerbound}(\alpha) &= m_{lower} \cdot \alpha + b_{lower} \end{aligned} \quad (3.25)$$

From now on, an entire curve family is cover by 9 coefficients 3.26. Figure 3.12 serves to visually illustrate the process of curve approximation once again.

$$[m_1, m_0, b_2, b_1, b_0, m_{upper}, b_{upper}, m_{lower}, b_{lower}] \quad (3.26)$$

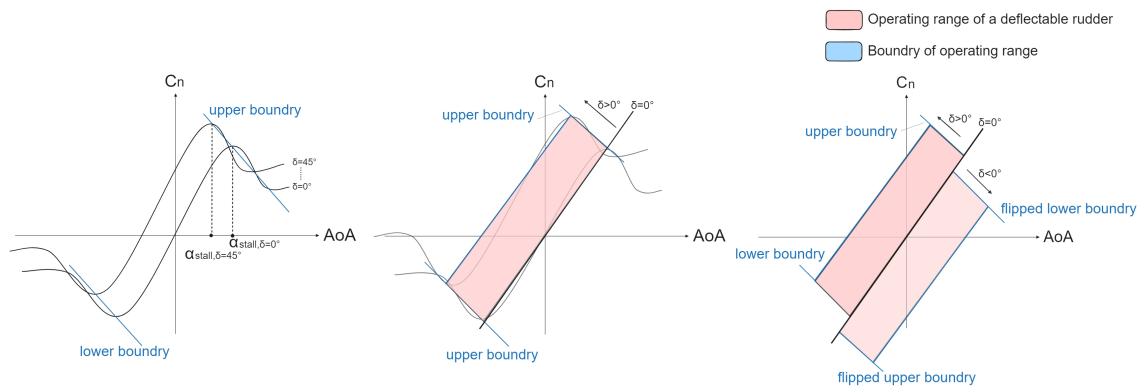


Figure 3.12: Approximation of C_n limited in laminar flow. The result is equivalent to the operating range of deflectable rudder's.

3.3.6 Response Outside the Operating Range

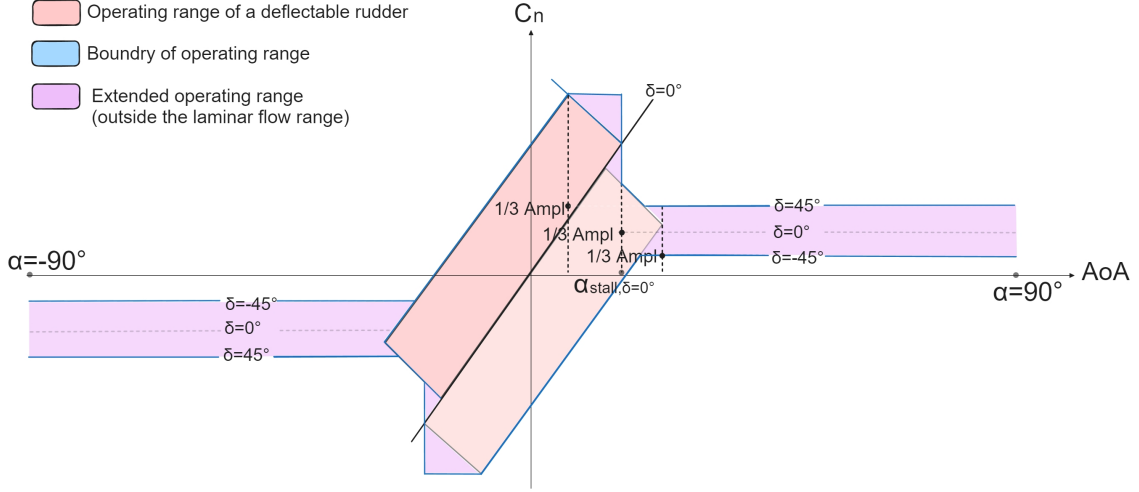


Figure 3.13: Extended operating area of C_n

The valid operational range for the rudder deflection discussed in the previous subsections are derived from the laminar range of the fin's control surface. There are basically two reasons why a operating point is not achievable: overwhelming and the occurrence of excessive AoA. In the case of overwhelming, the desired C_n value is above the maximum achievable C_n value. As described in the previous section, for this case, the next achievable value for C_n is returned, and the portion of control command which has not being implemented by the rudders, is forwarded to the control allocator of the side thrusters. The focus of this subsection is on the second case, where the parser does not find a corresponding operational range at the current AoA and airspeed. This case is figurely describable by drawing a vertical line through the current AoA on the diagram and this vertical line is not intersected with the area of the permitted laminar operating area. In this case, the controller for the rudder deflection is not able to respond appropriately at a prescribed operating point. For this reason, there is a need to extend the valid operating range beyond laminar range, even though the deflection of rudders no longer plays the essential role in the implementation of control commands in this stage.

At $\alpha = \alpha_{stall}$, the value for C_n and, therefore, the normal force F_n reach their maximum. Exceeding this AoA leads to an abrupt drop, which then stabilizes at only one-third of the maximum magnitude. Based on this observation, the operating range in the C_n -diagram is extended as follows: A vertical line is drawn, intersecting the curve at deflection $\delta = 0^\circ$; this corresponds an AoA being exactly the $\alpha_{stall, \delta=0}$. For each deflection other than zero, the stall AoA shows up earlier. In the case if current AoA lies among the range between

the corresponding α_{stall} and the $\alpha_{stall_{\delta=0}}$, the value $C_n = \alpha_{stall}$ is returned; if the AoA is beyond $\alpha_{stall_{\delta=0}}$, one-third of $C_n = \alpha_{stall}$ is returned as the result for the flight controller. This approach is applied up to $\alpha = 90^\circ$, the extended operating range in the C_n -diagram is shown in the figure 3.13.

3.3.7 Methodology for LookUp Table Retrival

The control allocator requires accompanying function algorithms for handling the described parameter list, whereby a parser is implemented for this purpose. This section explains the functional principle of the parser and the way to use the parameter list. The parameter list is a mapping of:

$$(V_\infty, \alpha, \delta) \mapsto C_n$$

where the factors δ and α is parameterized and is able to take any values as long as these two factors lies in the defined range, only the air speed is available as grid points in parameter list. It is possible that the query's airspeed corresponds exactly to one of the grid points in the parameter list; in practice, however, interpolation is necessary as the given airspeed more often lies between two grip points. The interpolation process is as follow: first, the parser looks for the right gap between two velocity grid points, within those the given airspeed is located, then the 9 coefficients of the corresponding grid points are called and for each of the grip points respectively, a C_n value is determined by the given α and δ ; in the last step, interpolate between these two C_n values. Figure ??

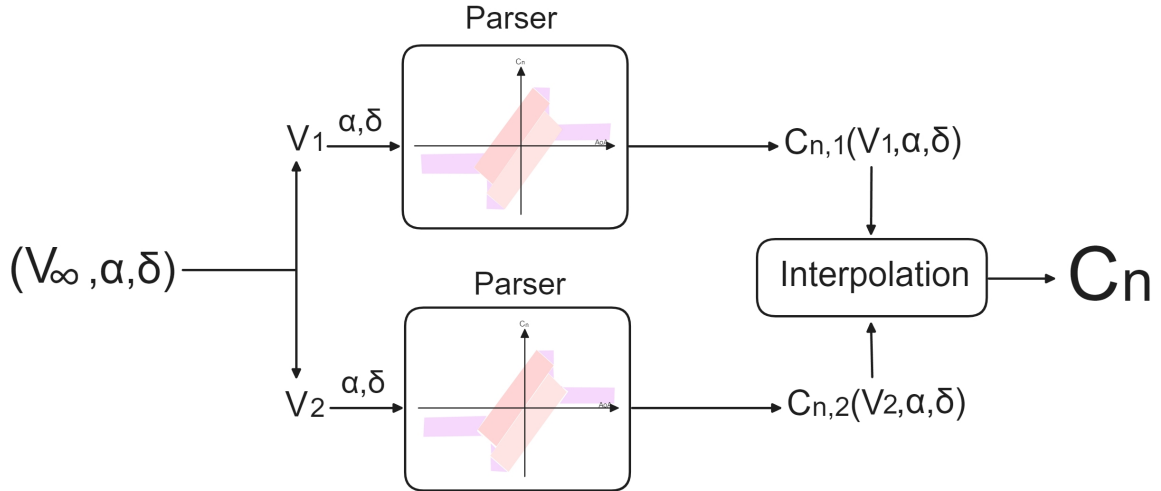


Figure 3.14: Looking for the value of C_n by given airspeed, AoA and rudder's deflection.

3 Methodology and Approach of nondimensional coefficient modelling

In the opposite case, the search for the rudder deflection δ via the parameter list by given set point C_n , current airspeed and AoA is also required. The mapping is as follows:

$$(V_\infty, \alpha, C_n) \mapsto \delta$$

This process is important for the fin's control allocator to determine which δ of deflection will provide the necessary amount of normal force to produce the desired torque for the airship. The target value of C_n is extracted from the desired angular velocity and passed to the parser along with the current air speed and AoA.

Thus far, due to point symmetry as demonstrated in 3.12, only half of the operating range, corresponding to positive deflection, has been considered, and this range has been approximated in the previous section using 9 coefficients. Now, the entire operating range of the rudder is to be completed by adding the other point-symmetric half, namely the range caused by negative deflection. Visually, the half with negative deflection complements the half with positive deflection by a 180-degree rotation, together with the upper- and lower limit boundary lines. Figure 3.15 demonstrates this workflow.

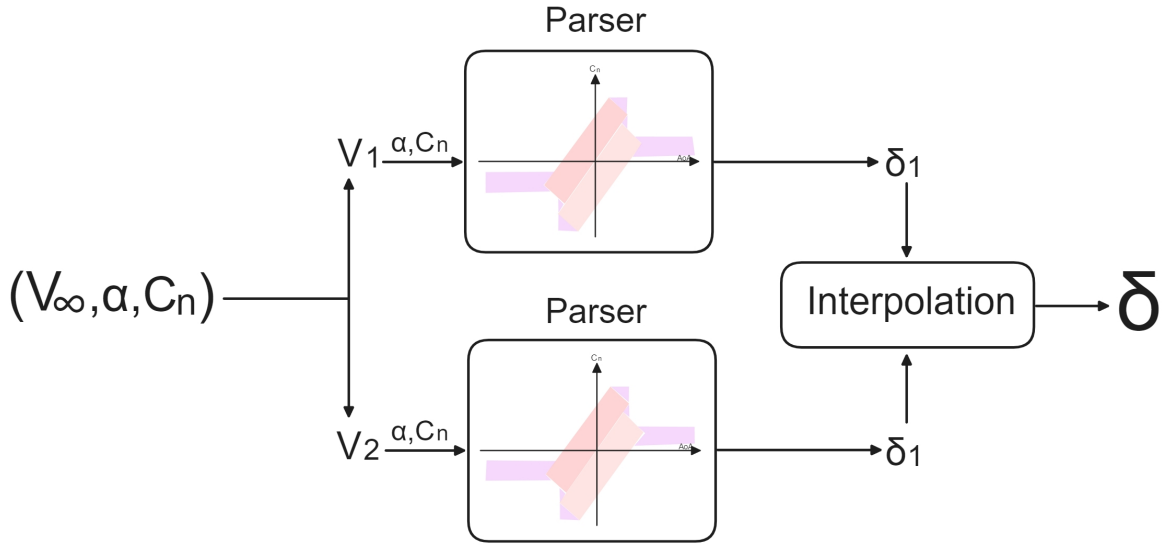


Figure 3.15: Looking for the appropriate rudder's deflection by given airspeed, AoA and rudder's deflection.

Similar to the previous mapping, the parser first checks the right gap, where the air speed value lies in between. Accordingly, the parser calls the curve-coefficients from the both grid points and each calculate a value for δ . It is possible that the demanding set point $C_{n_{sp}}$ is beyond the operating range of the rudder at current context, which underreaches the requirement despite deflecting to maximum 45 degrees to one of both sides, therefore,

3.3 Development of a Parameter List for Control Allocation

the parser checks at first in it's algorithm, whether the required C_n value is achievable in the current context. The maximum and minimum achievable value for C_n in the current context is calculated using the upper- and lower boundry curves. In a next step, the algorithm validates wheather the set point $C_{n_{sp}}$ lies among this range. In case of set point C_n lying outside of the operating range, the set point $C_{n_{sp}}$ is replaced by the limit value $C_{n_{lim,upper}}$ or $C_{n_{lim,lower}}$, respectively the upper or lower limit value. In the next step, the algorithm figures out the necessary deflection according to the set point $C_{n_{sp}}$ by converting the equation (3.26) to the search for δ , which is the equation (3.27). Only one of the both solutions δ lies in the ruder's valid deflection range, this δ is then forwarded to the outer side of the algorithm for interpolation with the result of the same process with another velocity grid point. The interpolation's result is the necessary deflection which is then forwarded to the deflection control.

$$C_{n_{sp}} = (m_1 \cdot \delta + m_0) \cdot \alpha + (b_2 \cdot \delta^2 + b_1 \cdot \delta + b_0)$$

$$\delta_{1,2} = \frac{-(m_1\alpha + b_1) \pm \sqrt{(m_1\alpha + b_1)^2 - 4b_2(m_0\alpha + b_0 - C_{n_{sp}})}}{2b_2} \quad (3.27)$$

3.3.8 Design and Integration of Rudder's Control Allocation

In the previous work by [3], a ready-made concept for the control allocator of the motors is designed and implemented, so that the control commands are assigned to the respective motors. This concept is extended by the control allocator for the rudders. Due to the reason in some situations the generated normal force does not completely cover the target value, the difference between the target- and realizable value is forwarded to the control allocator for the motors. The extended control system is shown in figure 3.16

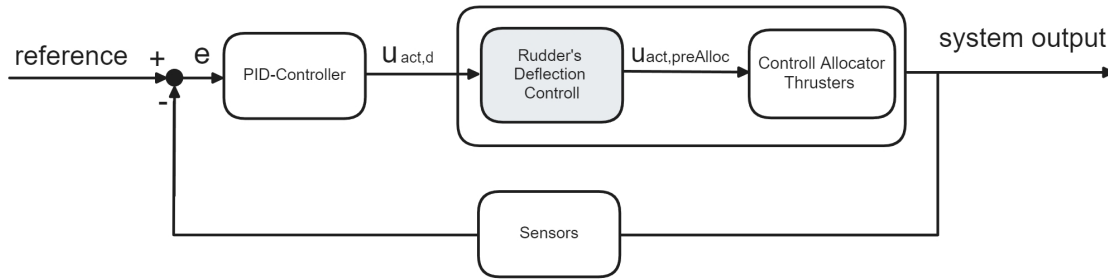


Figure 3.16: The system's control loop which is extended by a rudder's deflection controller, which is placed before the control allocator for the thrusters.

3 Methodology and Approach of nondimensional coefficient modelling

The system requires the desired angular velocity $\dot{\cdot} \cdot \underline{\Omega}_d$, which is represented in a body fixed Cartesian coordinate system with origin in the airship's CoV. The airspeed of i-th fin V_{∞_i} is calculated by using the equation (2.1) and (2.2). The fin's AoA is determined by applying a trigonometric function to the x- and y- components of the fin's airspeed:

$$\begin{aligned} V_{\infty_i} &= [v_x \quad v_y \quad v_z] \\ \alpha_i &= \arctan\left(\frac{-v_z}{v_x}\right) \end{aligned} \quad (3.28)$$

Firstly, the rudder's control allocator calculates the minimum and maximum effectiveness matrix $\underline{\underline{B}}_{\max}$ and $\underline{\underline{B}}_{\min}$ for all four rudders from the current context. An effectiveness matrix $\underline{\underline{B}}$ is a 3×4 matrix with each column's entry being the vector's realizable angular velocity in x , y and z and it is the difference between the $\underline{\underline{B}}_{\max}$ and $\underline{\underline{B}}_{\min}$:

$$\underline{\underline{B}}_{\text{act}} = \underline{\underline{B}}_{\text{act}_{\max}} - \underline{\underline{B}}_{\text{act}_{\min}} = \begin{bmatrix} b_{x,\text{fin}_1} & b_{x,\text{fin}_2} & b_{x,\text{fin}_3} & b_{x,\text{fin}_4} \\ b_{y,\text{fin}_1} & b_{y,\text{fin}_2} & \dots & \\ b_{z,\text{fin}_1} & \dots & & \end{bmatrix} \quad (3.29)$$

For the fin's control allocator, the dynamic is equal to the desired angular acceleration $\dot{\omega}_d$, hence the Newton-Euler-Equation is given by equation 2.7 and control command $\underline{u}_{\text{fin}}$ is aquired by applying pseudo inverse of the effectiveness matrix, which is then:

$$\underline{u}_{\text{act},\text{fin}} = \underline{\underline{B}}_{\text{act}}^{-1} \cdot \dot{\omega}_d \quad (3.30)$$

3.4 Establishing and Configuration of the Simulation Environment

The developed aerodynamic model and rudder's deflection control needs to be verified, for which the simulation tool Gazebo is being used. Gazebo is an open-source simulation commonly utilized for robotics and aerospace applications. The software provides a realistic physics engine and is a tool frequently used for testing and validation control systems in dynamic environments. It enables users to simulate the behaviour of robots, vehicles and other systems in 3D environments with real-time physics, hence it is widely used tools for control and sensor data handling, developing, testing and refining control algorithms prior before the model is deployed in real-world scenarios. This section is about the required implementation work in Gazebo.

3.4 Establishing and Configuration of the Simulation Environment

The simulation environment requires a model to simulate the reference environment. The analysing results from Xflr5 serves again for this purpose. In this term, the AoA of the $C_l\alpha$ and $C_d\alpha$ curves are extended to ± 90 degrees in Xflr5's "Batch Analysing" in order to cover the whole field of AoA. Similiar to the previous stage, the simulaton uses handles with normal- and axial forces either, hence a transformation from C_lC_d to C_nC_a is applied for the extended curves. For Curves that do not make it to ± 90 degrees, due to convergence problems in the numerical calculation, are artificially extended in post-processing. Following the unproven assumption that the resultant of normal and axial forces is maintained at very large AoA up to 90 degrees, but where the axial force goes to zero at $\alpha=90$, a straight line is taken to connect the last points at both ends of the C_a -curve respectively to one of the point at $C_a(\pm 90)$. The C_n curve is then determined by Pythagoras' theorem from last resultant value and values from extended C_a -curve.

4 Evaluation

This chapter validates and evaluates the methods and procedures used in the study. It begins with the evaluation of the Xflr5's simulation results, followed by a detailed discussion of the feasibility of the methodology and the evaluation and interpretation of the results from the simulation environment. The goal is to critically analyze the effectiveness and applicability of the approaches used and to provide insights into their strengths and potential limitations. In this way, the chapter aims to provide a comprehensive understanding of the overall performance and reliability in the simulation environment.

4.1 Interpretation and Evaluation of Xflr5 Results

The results of Xflr5 are an important foundation in this work, as the methods and procedures used are developed with reference to them.

4.1.1 Symmetry between FlapUp and FlapDown curves

Upon closer observation, it becomes evident that a point of symmetry exists between two C_l -curves with the same deflection angle δ , yet with reversed deflection direction. This point symmetry can be identified by the rounding of the amplitudes. The flap-up curve exhibit's a pronounced rounding in the lower amplitude and a narrow rounding in the upper amplitude. In contrast, the flap-down curves display the opposite behavior, with this phenomenon becoming increasingly evident as δ increases. Furthermore, irregularities in the curves, such as a spike or deviation from linearity in the laminar region, are consistently reflected in the other curves in a point-mirrored manner. This observation leads to the assumption that the flap-up curves are the point-mirrored curves of the flap-down curves, which results in a major simplification, as the curve to be considered is reduced by half:

$$C_l(\alpha, -\delta) = -C_l(-\alpha, -\delta) \quad (4.1)$$

The C_d curves within the laminar range exhibit a parabolic shape with an upward opening.

In regions outside of the laminar range, the increase in deflection is less pronounced. As the rudder deflection increases, the opening narrows, and in the region around the origin, the curve appears with increasing fluctuation and a slight upward shift. In a C_d -curve, there is a distinct symmetry between the FlapUp and FlapDown curves with equal deflection, with the curves mirroring each other around the y -axis:

$$C_d(\alpha) = C_d(-\alpha) \quad (4.2)$$

4.1.2 Aerodynamic in low speed range

The speed range from 0 to 60 km/h is replaced by 10 discrete values, resulting in the formation of 10 classes of curves. The characteristics of curves belonging to different classes are examined and compared in this subsection.

Initially, the neutral C_n -curves ($\delta = 0$) from various airspeed classes are compared, since the non-neutral curves are just the displacement of the neutral curves. The amplitude's magnitude of the C_n curves increase and α_{stall} appears later when airspeed and Reynolds Number gain. This indicates that in case of higher airspeed, the fin experiences greater lift and the range for laminar flow expands. Consequently, the operating range of the fins also expands at higher speeds.

In case of a very low air speed $V_\infty < 5$ km/h and thus a Reynold's number below 44000, the C_n -curves first show a local maximum and then a local minimum around the origin, only then do the curves behave monotonically increasing until the appearance of stall. This phenomenon only occurs on curves with a very low air speed and therefore low Reynold's numbers. To observe the curve behavior more closely, additional curves are generated from the speed range with Xflr5. At speeds below 3 km/h, the laminar and turbulent regions can no longer be clearly distinguished from each other. On the other hand, the occurrence of local minimum and maximum around the origin only disappears from an air speed of 7km/h and from this point onwards, the C_l -curve behaves almost linearly as with curves at higher speeds. The C_d -curves in low speed range show a v-shaped form in contrast to the curves of higher speed range, which have a u-shaped form and vary only slightly from each other. This property again confirms the observation that the transition between laminar and turbulent regions is no longer clearly recognizable in the low velocity range. This raises a concern about the linear approximation and interpolation of C_n -curves in the low velocity range, which results from a coordinate transformation of C_l and C_d .

4 Evaluation

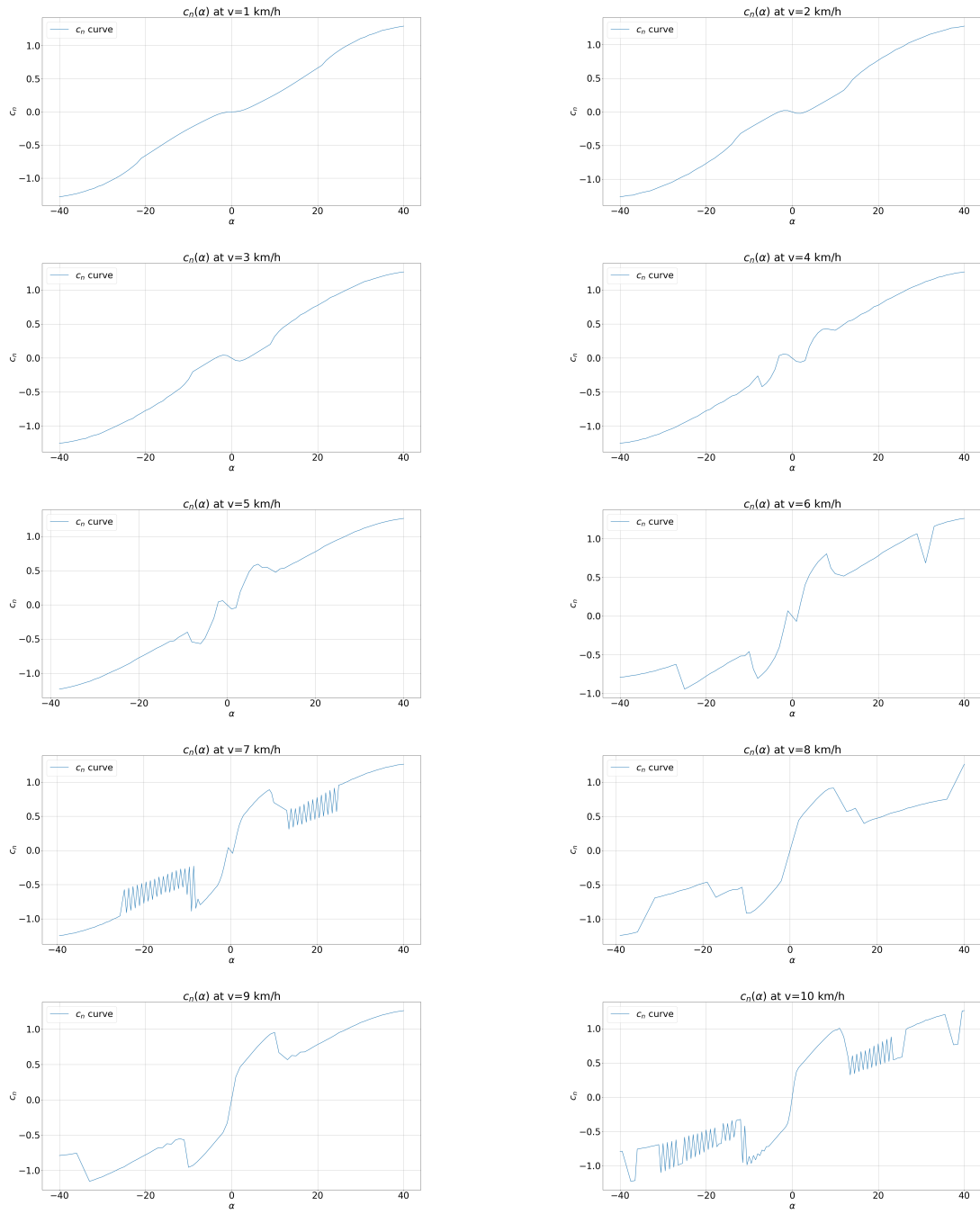


Figure 4.1: curve family of $C_n(\alpha)$ for airspeed under 10km/h

4.1 Interpretation and Evaluation of Xflr5 Results

In order to examine the behavior of C_n -curves in the lower speed range into the details, C_n -values of the speeds from 1 to 10km/h, C_n -curves are generated again via C_l and C_d in Xflr5. The resulting curves are displayed in figure 4.1. At the speed of $V_\infty = 1$ km/h, the curve is monotonically increasing. For V_∞ around 4 to 7 km/h, each curve demonstrates a double change of direction around the origin. This change of direction is discernible at $V_\infty = 7$ km/h and occurs exclusively for the linear approximation between the two stall angles. From $V_\infty = 8$ km/h, the phenomenon is no longer discernible. In light of this observation, the first curve class is elevated from $V_\infty = 5$ km/h to $V_\infty = 7$ km/h. Between $V_\infty = 0$ and 7 km/h, linear interpolation remains the method of choice, although the interpolation result may diverge more from the curve in this range.

4.1.3 Evaluation of Approximation of C_n

The parameterization of C_n -curves is used for further implementation and has an impact on the subsequent results. It is therefore important to evaluate the Approximation of C_n .

The C_n -curve is parameterized using a first-order-polynomial (linear straight line). The influence of the deflection is parameterized by a second-order-polynomial, since the distances between the C_n -curves within a curve family is getting smaller when the deflection increases. In practice, this means that the deflection does not have a proportional effect on the normal force, but increasingly loses impact as the deflection increases. On the other side, using higher order polynomial leads to more complexity when solving the polynomial equation. The second order polynomial fits the curve and is simple to solve as well. Since the shifted curves within the laminar region exhibit strong parallelism, the slopes of the curves are very similar. Therefore, a linear equation is sufficient to approximate the slope m . In the lower part of figure 4.2, the approximated coefficients are presented. It is evident that the chosen order achieves a very good approximation.

4 Evaluation

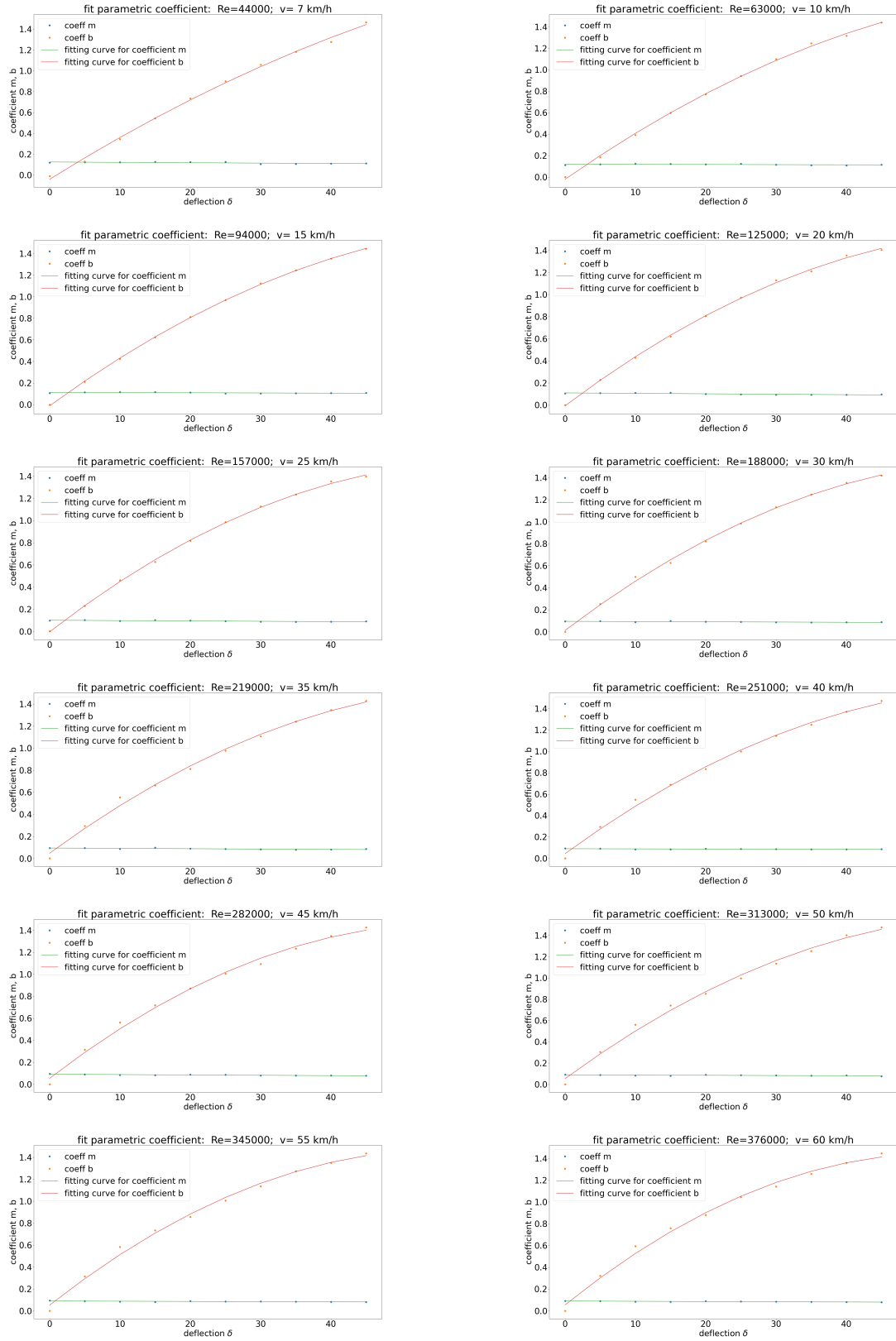


Figure 4.2: Fit parametric coefficients of $C_n(\alpha)$, where the slop m is fitted using 1st-order polynomial and the offset b is fitted using 2nc order polynomial

4.2 Validation of the Implementation in PX4

In order to assess the performance of the rudder's deflection control in PX4 flight controller, a comparison between the simulation results from Gazebo and the log data generated by PX4 was conducted. This comparison aims to identify discrepancies between the expected behavior (as perceived by PX4) and the actual outcomes observed in the simulation environment. By analyzing these differences, the accuracy of the control algorithms is evaluated and the system's overall response is refined, ensuring that the control system behaves as intended under real-world conditions.

The focus is on the normal force coefficient C_n . In PX4, C_n is determined by reconstructing an approximated C_n -curve, where the coefficients are obtained from the parameter list. In contrast, Gazebo uses raw functional data for C_n in the simulation. This difference in approach is crucial for understanding the discrepancy between the approximated C_n in PX4 and the "real" raw C_n values in the same given conditions (AoA, airspeed and deflection). This analysis helps to assess how well the approximation represents the actual behavior under similar scenarios. Thus, a direct comparison of the normal force coefficient (C_n) as calculated by PX4 and as simulated in Gazebo is presented, with one diagram dedicated to each fin. The diagrams are shown in Figure 4.3.

The comparison of the curves from PX4 and Gazebo reveals some discrepancies; however, these differences are within an acceptable range. The overall trends of both temporal progression align quite well, as they exhibit simultaneous peaks and valleys. These peaks correspond to the turning action of the airship at one of the waypoints. Apart of these peaks, the temporal progression from the Gazebo simulation remains relatively stable and close to the zero line, while the PX4's temporal progression deviates from the zero line more noticeably. This deviation suggests that PX4's fitting curve for C_n introduce minor fluctuations when compared to the raw C_n data, but the general behavior remains consistent between both systems.

The normal force F_n , derived from the normal force coefficient (C_n), is also taken into closer study. The temporal progression of F_n are demonstrated in Figure 4.4. Both temporal progression from PX4 and Gazebo shows clear peaks at the same times, indicating that both systems detect the airship's response during key moments of directional change. However, the PX4 curve occasionally overshoots or undershoots compared to Gazebo, and the magnitude of the differences between the two data sets is more pronounced in some segments, in comparison to the datasets of C_n . In calm phases, the Gazebo data maintains a more stable profile around zero, while the PX4 curve shows more variation, though this difference remains within acceptable limits for the approximation method.

4 Evaluation

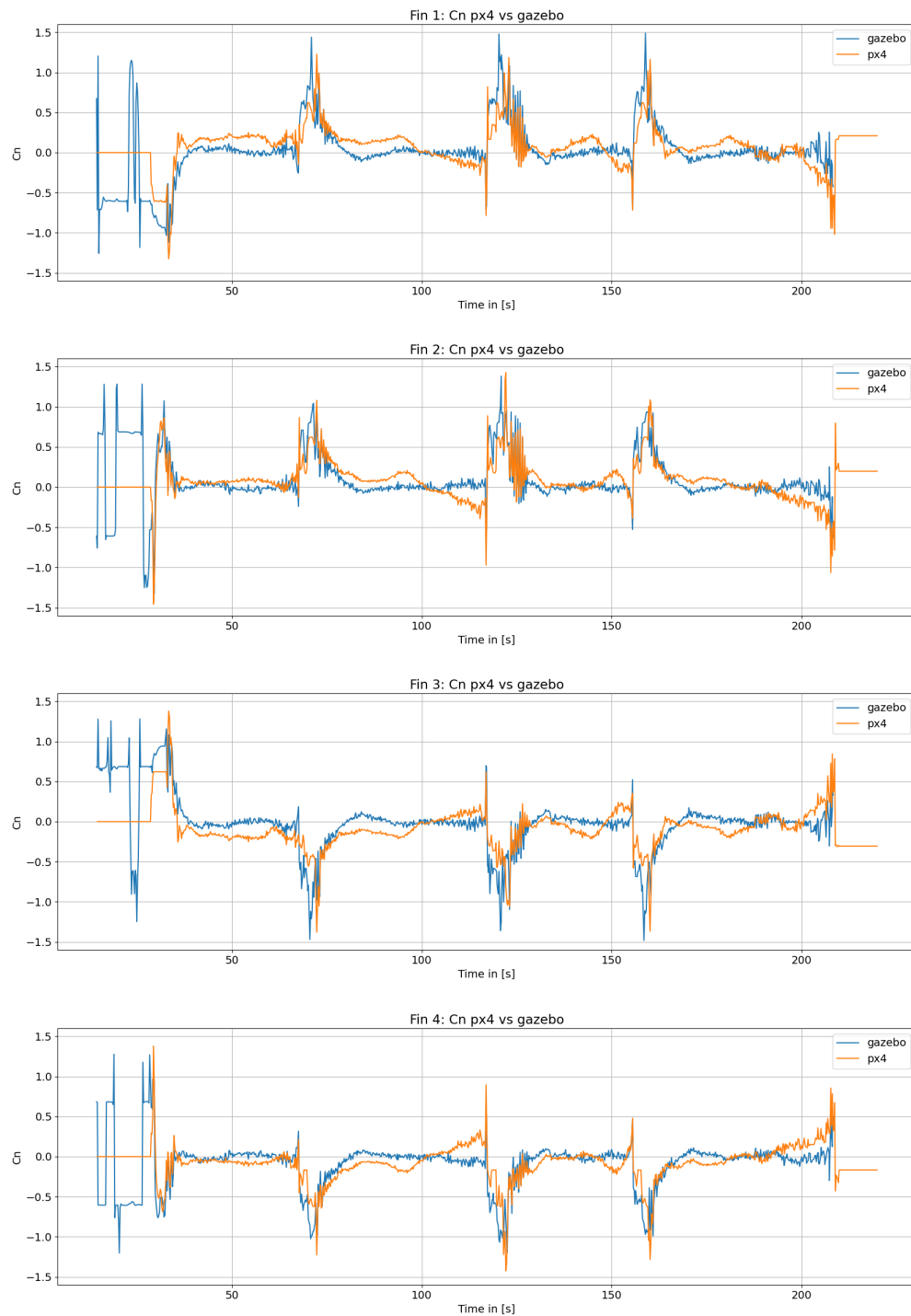


Figure 4.3: Comparison of C_n between Px4 and Gazebo respectively to each fin.

4.2 Validation of the Implementation in PX4

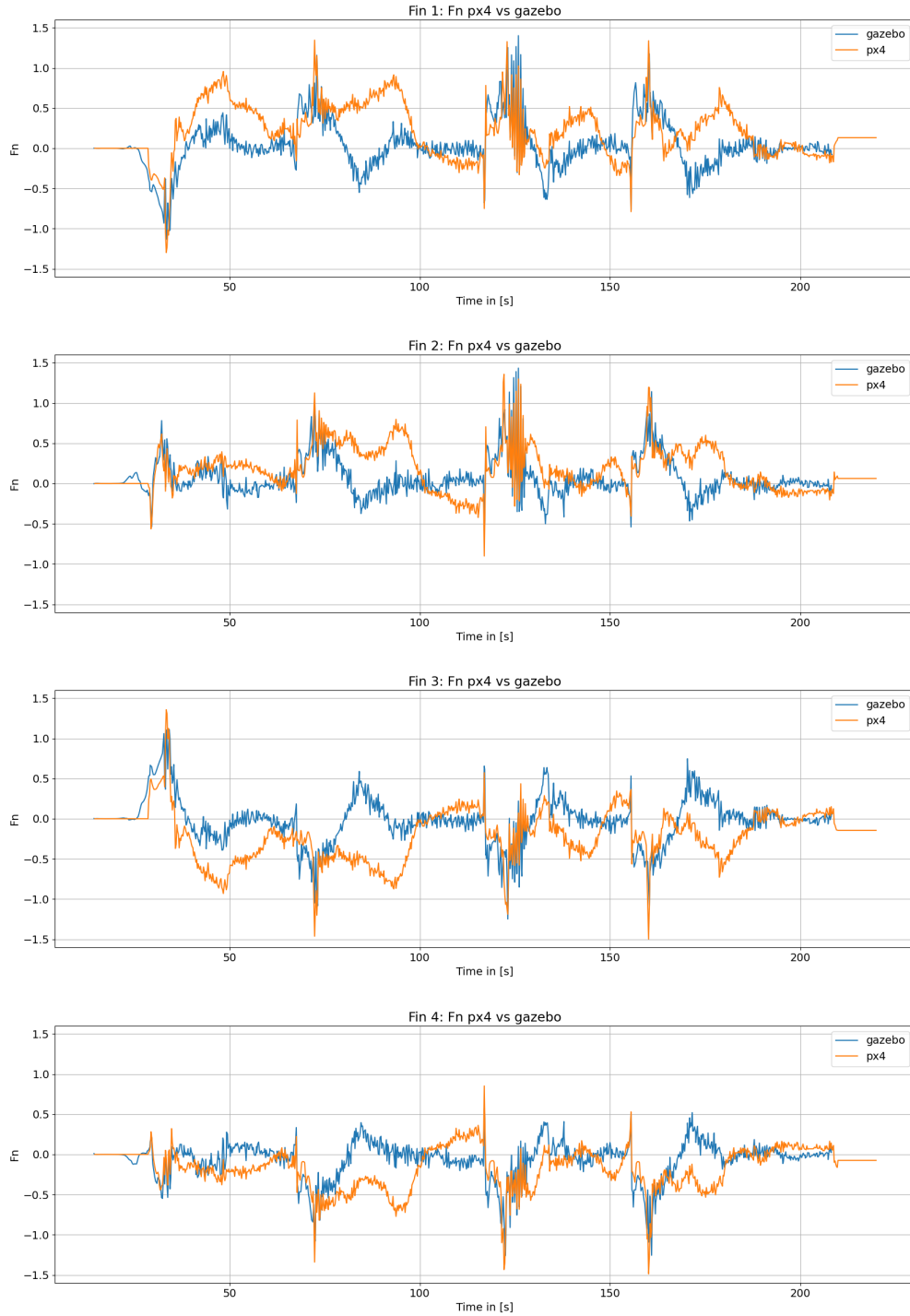


Figure 4.4: Comparison of F_n between Px4 and Gazebo respectively to each fin.

4.3 Proof of the Negligibility of Coefficient C_a

In Chapter 3, the assumption is made that the non-dimensional coefficient C_a is sufficiently small compared to C_n to be considered negligible. This assumption serves as a key requirement for simplifying the aerodynamic model. In this section, the relationship between C_n and C_a is examined in detail to provide a justification for this assumption. By closely analyzing the magnitude of both coefficients, a quantitative proof is offered to validate the neglect of C_a , thereby reinforcing the importance of this assumption in the model.

In the diagram series in Figure 4.5 and 4.6, the temporal progression of the non-dimensional coefficients C_n and C_a from the simulation in Gazebo is shown. It is clearly visible that C_a remains close to 0 throughout the entire course when compared to C_n . Only during periods of relatively high normal force F_n does the axial non-dimensional coefficient C_a , and thus the axial force F_a , become noticeable but only for a very brief period, although the magnitude of C_a never reaches $\frac{1}{3}$ of the magnitude of C_n . This observation serves as evidence that C_a is neglectable.

4.3 Proof of the Negligibility of Coefficient C_a

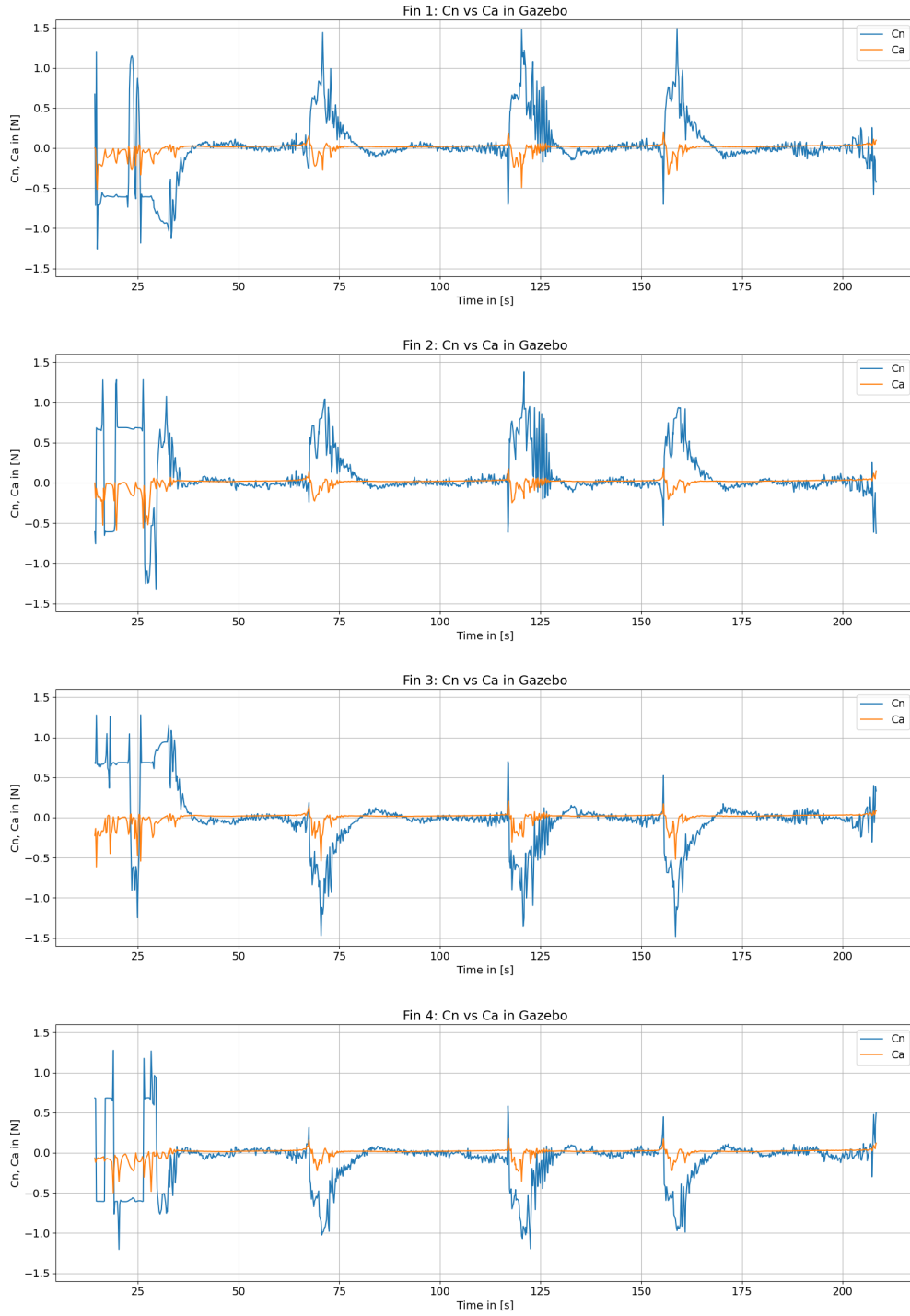


Figure 4.5: Comparison between C_n and C_a of each fin.

4 Evaluation

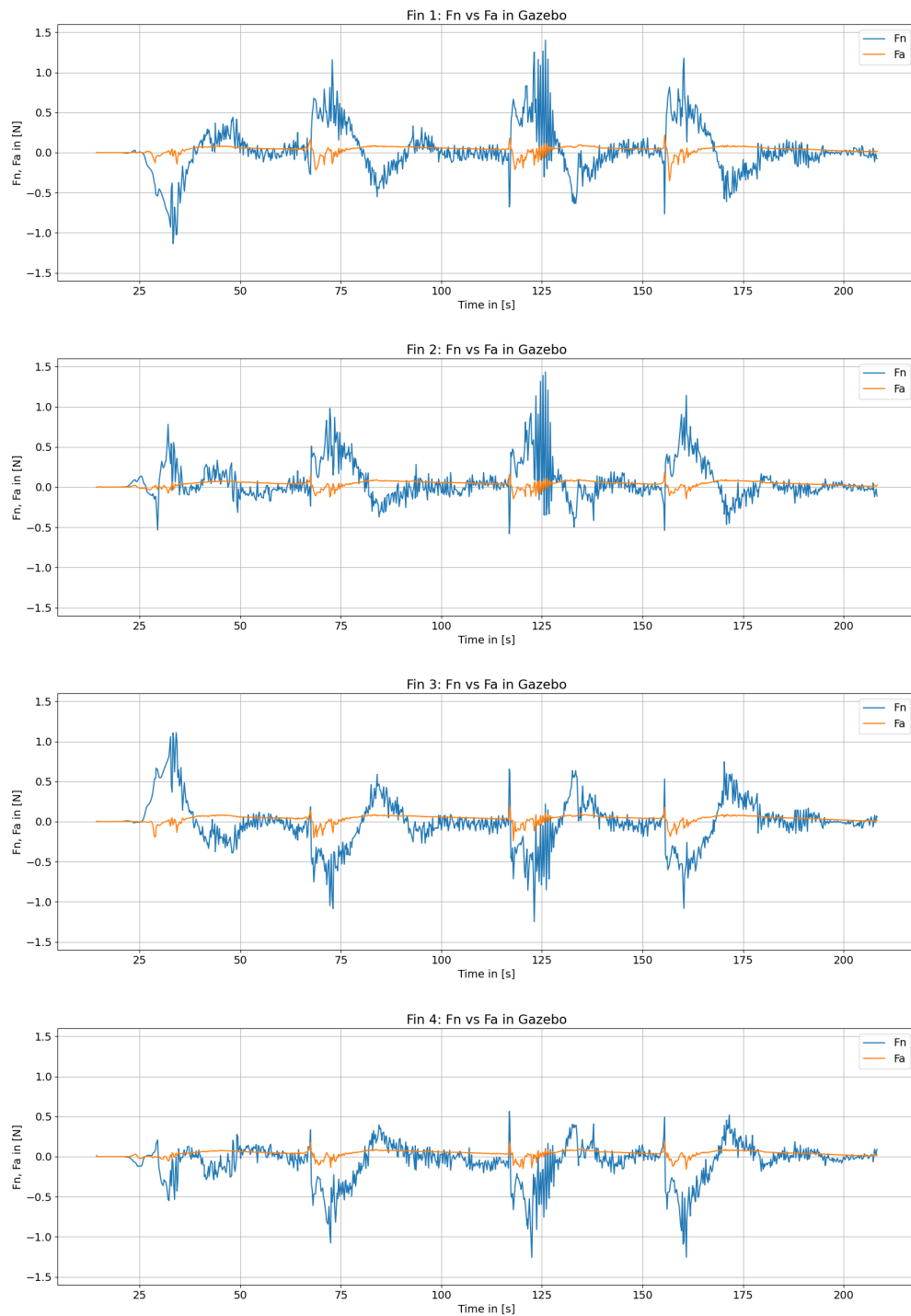


Figure 4.6: Comparison between F_n and F_a of each fin

4.4 Evaluation of flight performance

The airship is validated in the simulation in order to test the functionality of the rudder's deflection control and the effectiveness with and without cooperation with the allocated side thrusters.

To validate this, the airship should fly along a given route defined by a sequence of waypoints. First, the airship rises from the initial starting point to an altitude of 10 meters, which is called the first waypoint w_1 ; then five more waypoints ($w_1...w_5$) follow, forming a circle. The waypoints are of the same altitude, so there is no need to climb or descend when following the waypoints. The coordinates of the waypoints are given in the inertial frame by following:

$$w_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad w_1 = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix} \quad w_2 = \begin{pmatrix} 0 \\ 14.65 \\ 10 \end{pmatrix} \quad w_3 = \begin{pmatrix} 14.6 \\ 27.7 \\ 10 \end{pmatrix} \quad w_4 = \begin{pmatrix} 26.6 \\ 26.7 \\ 10 \end{pmatrix} \quad w_5 = \begin{pmatrix} 1.6 \\ 1 \\ 10 \end{pmatrix}$$

It is required to travel as straight as possible between the waypoints, while altitude is maintained as much as possible. The maximum air speed is limited to 10 km/h and the Wind is excluded by setting the wind speed to 0km/h in the simulation, as the rudder's deflection only depends on the AoA and the x - and z -component of the relative velocity of each fin. With crosswinds, the system would not be able to response correctly, which is indeed part of the topic "path planning" and is outside the scope of this proposal. The airship's movement is then observed and operational data from PX4 and Gazebo are recorded. In the evaluation, the airship's flight behavior is analyzed by comparing the data from PX4 and Gazebo.

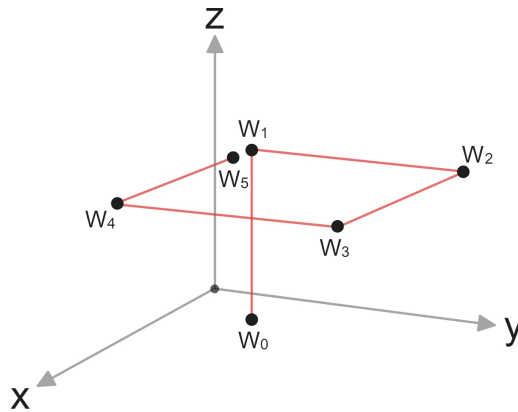


Figure 4.7: Predefined trajectory waypoints in the simulation for guiding the airship

4.5 Trajectory comparison

As described in Figure 3.16 from the previous Chapter, the control command is forwarded to the rudder deflection and control allocation of the side thrusters, which are connected in series. The control command $\underline{u}_{act}$ is initially directed to the rudder's control. However, if understeering occurs and the control command $\underline{u}_{act}$ is not fully executed by the rudder's deflection, the remaining control command $\underline{u}_r$ is forwarded to the side thrusters to complete the control action. In order to figure out the effectiveness of the rudder's deflection in comparison to the side thrusters, the described route is flown in two variants: first with both rudder's deflection control and control allocation of the side thrusters; the second one without rudder's deflection control but only with side thrusters. The trajectories are compared.

The trajectories from the top view are shown in Figure 4.8. In the respective runs, the airship turns very skilfully around the waypoints and then corrects itself very quickly back onto a straight course. The trajectories are very close to each other. This extremely high similarity of the travelled trajectories is a sign that the rudder's deflection control works as required and gets on well with approach with only control allocation of the side thrusters. The altitude of the individual sections are plotted in the diagram. A small oscillation is recognisable on the diagram, but the fluctuation is less than 0.5m and the altitude of 10m is maintained throughout the entire flight from waypoint 1 to 5. Overall, the airship is very good at maintaining altitude when turning around the corner by ruder's deflection control.

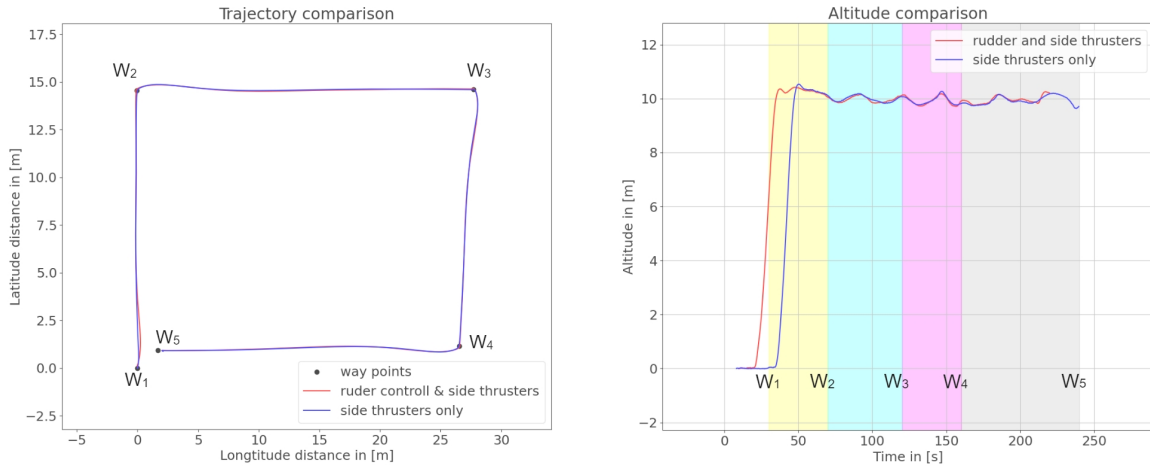


Figure 4.8: Comparison of the recorded trajectory and altitudes from both runs in Gazebo simulation

4.5 Trajectory comparison

In order to find out how far the rudders implement the desired control command $\underline{u}_{act,d}$, the allocated torque is examined in more detail. The allocated torque is a vector with 3 components, each represents the torque around x -, y - and z - axis. The comparison of the torques is presented in Figure 4.9 and 4.10. Figure 4.9 shows the allocated torques resulting from the combination of rudder's deflection control and side thrusters respectively in x -, y -, and z -direction; Figure 4.10 shows the allocated torques for only using side thrusters. The main focus is on comparing the torques around the y -, and z between these two concepts.

Apart from the ascent phase, four distinct groups of peaks can be seen in each diagram. This occurs when the airship reaches a waypoint and makes a turn for directional change, requiring significant torque generated both by the rudder deflection and the side thrusters. When using only the side thrusters, the motors continuously produce torque around the y -, and z -axes. However, when both the rudder and side thrusters are engaged, the side thrusters operate only briefly (for less than 3 seconds) during the turn, and no torque from the side thrusters is visible for the rest of the time. Furthermore, the peak torque generated by the side thrusters is smaller when used in combination with the rudder, compared to their use alone. This suggests that the rudder's deflection control handles most of the workload. Only during sharp turns and when the current airspeed is insufficient to provide adequate effectiveness for the fins and deflecting rudders, there a short-term need for support from the side thrusters. It is worth mentioning that the route planner currently in use reduces the airship's flight speed before each directional change to avoid unnecessary drifting and rolling. Although the speed reduction before a directional change enhances the stability in movement, it diminishes the overall aerodynamic effectiveness of the fins and rudder as well. This requires reconsidering and optimization of the airship's route planner, with a focus on the aerodynamic efficiency of the fins and rudders. However, this step goes beyond the goal of this proposal.

4 Evaluation

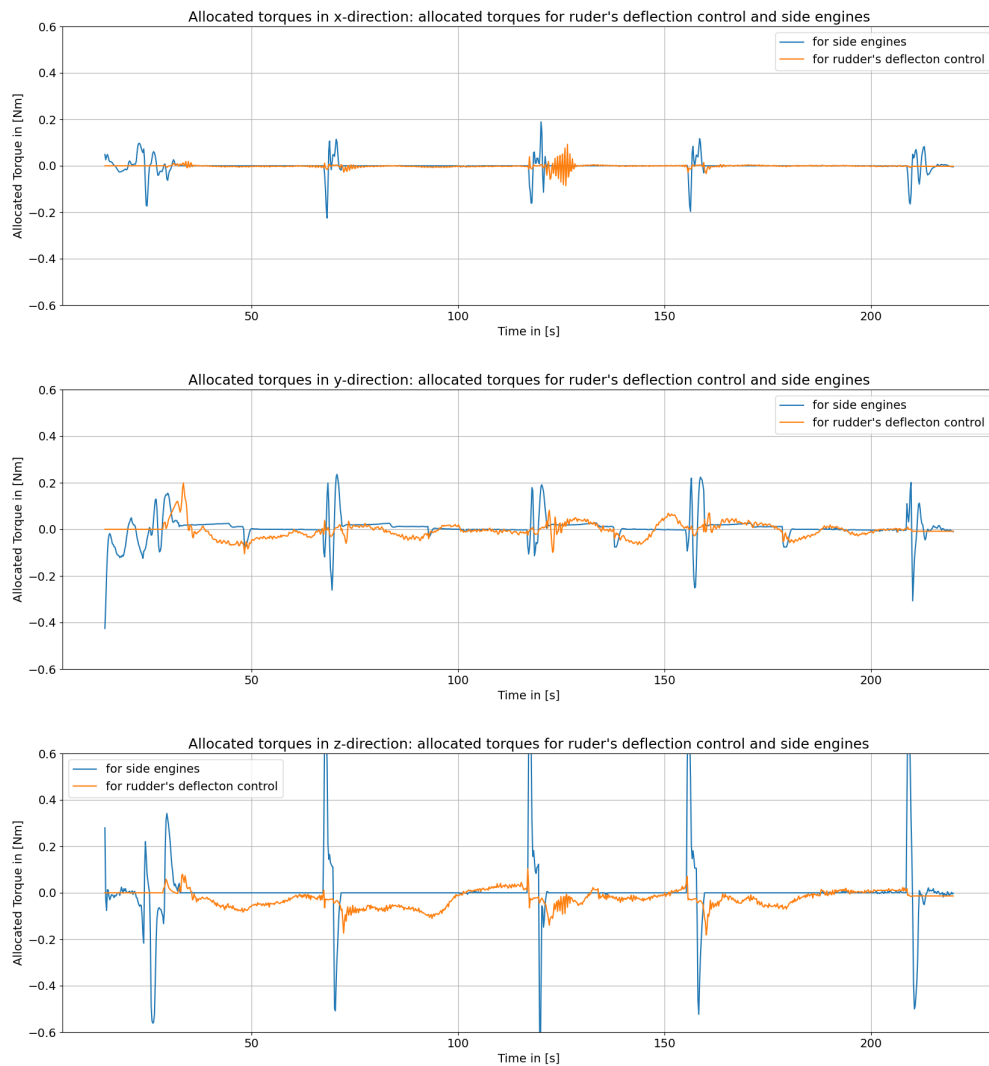


Figure 4.9: Comparison of allocating torques for the run with both side thrusters and rudder deflecton

4.5 Trajectory comparison

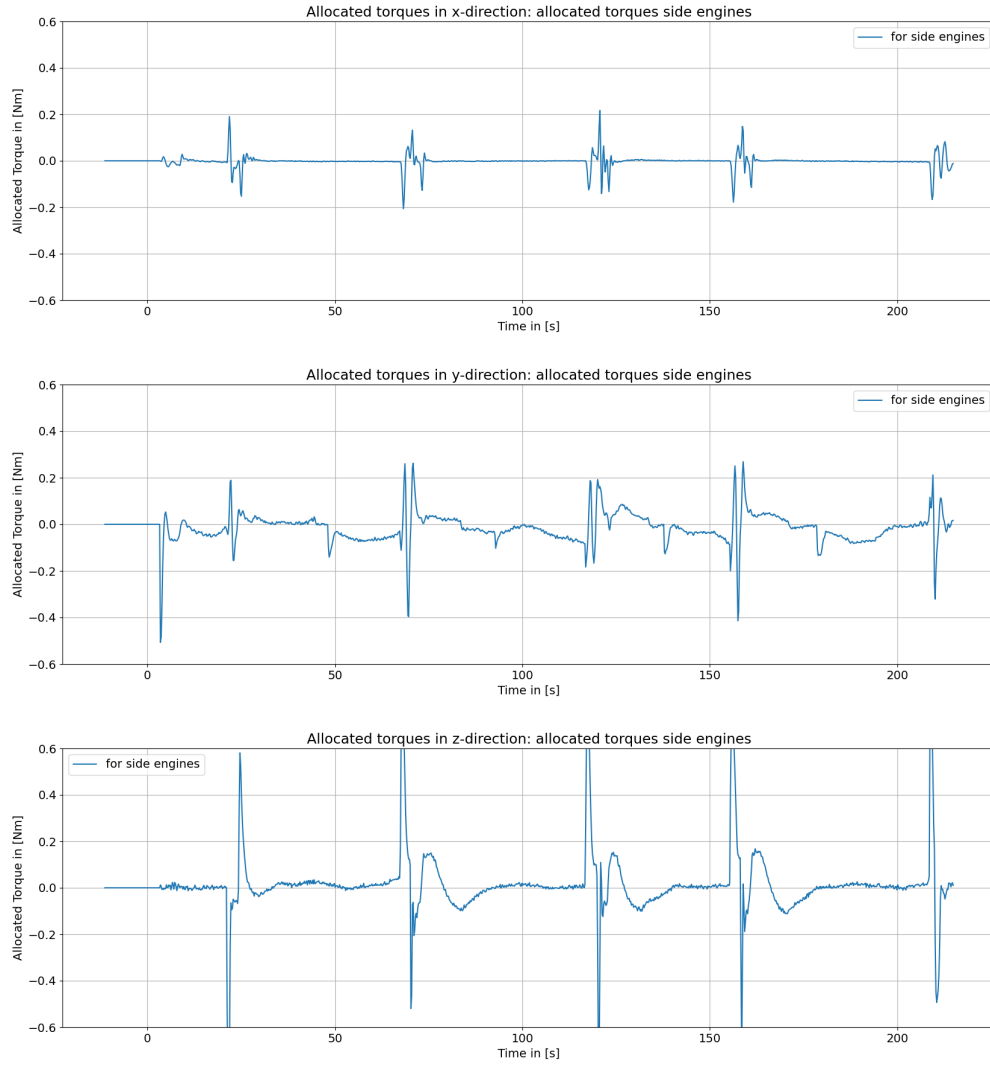


Figure 4.10: Comparison of allocating torques for the run with only side thrusters

5 Conclusion

This master's thesis focuses on the aerodynamic modeling and control system design for an omnidirectional airship with deflectable fins, aiming to optimize energy efficiency and enhance maneuverability. It builds upon result from [3] and extends the existing control system by incorporating rudder deflection control. The objective is to reduce the load on the side thrusters using aerodynamic forces, thereby optimizing the airship's energy efficiency.

In first step of this proposal, the aerodynamic properties of the fins based on the NACA 0015 airfoil are examined. The focus is on modeling the relationship between AoA α , relative airspeed V_∞ , rudder deflection δ and aerodynamic forces such as lift L , drag D , normal F_n , and axial force F_a , where the forces primarily represented by their non-dimensional coefficients. The modeling of these coefficients is approached in two ways: analytically and numerically. The numerical approach proves to be more superior in terms of plausibility and feasibility, thus the numerical approach serves as the fundamental basis for modeling of the non-dimensional coefficients and further solution strategies.

By utilizing the numerical CFD program XFLR5, key non-dimensional coefficients such as lift, drag, and moment were modeled to understand their behavior under various angles of attack (AoA) and rudder deflections. As a result, the aerodynamic force coefficients are expressed as functions of AoA and rudder deflection δ , represented by families of curves of individual airspeed categoration. Each family corresponds to the operational range of the rudder deflection. For simplicity, lift and drag are transformed into normal and axial forces through a rotational transformation around the AoA. The normal force is always perpendicular to the fin's surface, while the axial force is parallel to the airfoil's chord, though the axial force is considered negligible for simplification. Thus, the aerodynamic forces are reduced to a normal force, and the corresponding non-dimensional coefficient C_n , as a function of AoA α , is used as the aerodynamic reference for rudder deflection control. Simultaneously, corresponding algorithms were developed to facilitate value lookup for this relationship, which was then integrated into the rudder deflection control system. This control system determines the appropriate deflection for each fin based on the current AoA, airspeed, and setpoint value, forwarding any unmet steering commands to the control allocation for the side thrusters.

Subsequently, this setup was tested in the Gazebo simulation environment. The simula-

tion uses XFLR5 datasets as the aerodynamic reference, with the datasets extended to cover AoA values up to $\pm 90^\circ$, ensuring complete AoA coverage. In the final stage, simulations were conducted to validate the implemented algorithms and assess discrepancies between the flight controller’s decisions and the simulated environment. For this purpose, waypoints were set to form a rectangular flight course for the airship to follow. To demonstrate the effectiveness of the rudder deflection control, test flights were conducted in the simulation—once with both side thrusters and rudder deflection control, and once using only side thrusters without rudder control. The flight data were then compared. The results indicate that the fins, with their rudder deflection control, take over most of the load from the side thrusters. Only in specific situations, such as during sharp turns, does the system rely on brief support from the side thrusters.

The aerodynamic model requires further improvements, as certain simplifications were made in the approach, with some assumptions not yet thoroughly investigated. One area that still needs to be explored is the behavior and effectiveness of rudder deflection outside the laminar range. Although this range is not the focus of this thesis, it is still significant, as extending the analysis beyond the laminar region could broaden the operational scope of higher AoA. It remains uncertain how realistic the results provided by Xflr5 are outside the laminar range. The expansion of the C_n -curve was implemented only as a temporary solution to avoid edge cases during value lookup.

Another critical part of the thesis involved is the evaluation of the behavior of rudder’s deflection control and its impact on airship dynamics. By approximating the curves representing aerodynamic forces with straight lines, the complexity of the control system was reduced while maintaining accuracy in the laminar region. The analysis demonstrates that this approximation leads to a flight behavior which still corresponds to the raw aerodynamic data set referenced in the simulation. Although there is discrepancy between the flight controller and the simulation environment, the deviation is still in an acceptable range.

The approximated model becomes particularly critical during flights at very low speeds (below 7 km/h), where the fins, combined with their rudder deflection, do not provide sufficient effectiveness, thereby limiting their ability to meet the required performance. This issue is especially evident when navigating sharp turns, as the route planner significantly reduces the airship’s flight speed for stability purposes before entering the turn. This highlights the need to adjust the route planner’s algorithms to ensure that the reduction in speed is optimized in relation to the fins’ effectiveness.

Overall, the proposed control allocation method and aerodynamic model provide a solid foundation for enhancing the flight efficiency of airships under variable wind conditions and AoA. These findings pave the way for future optimizations and expansions, particularly in extending the operational AoA range and re-optimizing the route planner’s

5 Conclusion

algorithm to account for the aerodynamic effectiveness of the fins and rudders as a function of flight velocity.

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